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via 3D Semantic Scene Graph for Zero-shot Long-Horizon
Vision-Language Navigation

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DECLARATION OF ORIGINALITY

Title of Dissertation

Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

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Abstract

Vision-Language Navigation (VLN) aims to develop autonomous robotic agents capable of executing natural language instructions in unmapped environments. However, scaling these systems to Long-Horizon VLN (LH-VLN) introduces critical bottlenecks, primarily the semantic-spatial disconnect, exponential memory overload, and catastrophic error propagation across sequential sub-tasks. Existing modular architectures often rely on either unstructured historical logs or rigid 2D semantic maps, failing to seamlessly couple cross-task cognitive reasoning with dense 3D geometric execution.

To address these limitations, this dissertation proposes a novel Dual-Channel Memory Architecture for zero-shot LH-VLN. Inspired by biological cognitive structures, the framework explicitly decouples spatial and experiential memory. A dynamic, Hierarchical 3D Semantic Scene Graph (3DSG) functions as the spatial "Hippocampus", strictly maintaining the real-time geometric and semantic state of the environment. Concurrently, an independent Vectorized Experience Memory engine acts as the "Cerebral Cortex", actively extracting, storing, and pruning trial-and-error heuristics and failure traps for cross-task knowledge transfer. Furthermore, a map-free 4D Frontier Semantic Scoring mechanism and a Kinematic Safety Filter are introduced to eliminate blind exploration and prevent physical collisions.

The complete framework was evaluated on the Habitat simulation platform utilizing the large-scale LHPR-VLN benchmark (HM3D scenes). Despite operating under a strict single front-facing view constraint, the proposed method significantly outperformed existing zero-shot and finetuned baselines in maintaining sequential task consistency. The system achieved an Independent Success Rate (ISR) of 42.41%, a Conditional Success Rate (CSR) of 29.48%, and an Oracle Success Rate (OSR) of 71.43%. Comprehensive ablation studies validate that the integration of 3DSG spatial grounding and active experiential memory effectively mitigates cognitive hallucinations. Ultimately, this research provides a scalable and robust cognitive substrate for autonomous embodied agents executing complex, lifelong tasks in unconstrained real-world settings.

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Nomenclature

Symbols

Symbol	Description
α, β	Weight coefficients for Information Gain (IG) and Room Unexplored Ratio (UR)
$\gamma_{room}, \gamma_{obj}$	Weight coefficients for room-level and object-level semantic similarity
δ	Distance penalty coefficient in frontier evaluation
θ_{conf}	Confidence threshold (e.g., 0.6) for deterministic semantic routing
D_{th}	Distance threshold for evaluating navigation success (e.g., 1.0m or 3.0m)
τ	Time constant (half-life) for exponential decay of memory
N_{max}	Maximum memory capacity before active pruning
E_i	Discrete experience memory node
V_i, V_q	High-dimensional semantic text embedding vector
o_i	Execution outcome of a logged navigation phase (Success/Failure/Exploration)
γ_i	Spatial anchor identifier mapped to a 3DSG vertex
ψ	Text embedding transformation function
CE_i	Contribution Expectation index for memory pruning
$\Omega(E_i)$	Pitfall mitigation modifier for trap avoidance
w_{rel}, w_{reach}	Weight scaling coefficients for memory relevance and reachability
S_t	Multi-modal state payload at decision interval t
I_{global}	Global natural language instruction string
$s_{current}$	Active sub-task dictionary tracking the target text
G_t	Active directed topological scene graph at step t
X_t	Immediate state string of the agent (position and room)
C_{exp}	Historical experience context
$P_{snap}, P_{retract}$	Snapped navigable target point and retracted safe target point
$F_{unvisited}$	Set of currently unvisited frontiers
R_{all}, O_{all}	Sets of all known regions and known objects
P_{agent}	Current agent position in the physical coordinate frame

Acronyms

Acronym	Description
VLM	Vision-Language Model
VLN	Vision-Language Navigation
LH-VLN	Long-Horizon Vision-Language Navigation

LHPR-VLN	Long-Horizon Planning and Reasoning in VLN benchmark
3DSG	Hierarchical 3D Semantic Scene Graph
TSDf	Truncated Signed Distance Function
SR	Success Rate
OSR	Oracle Success Rate
ISR	Independent Success Rate
CSR	Conditional Success Rate
CGT	CSR weighted by Ground Truth
SPL	Success weighted by Path Length
NE	Navigation Error
IG	Information Gain
UR	Unexplored Ratio
LLM	Large Language Model
RGB-D	Red-Green-Blue plus Depth
GT	Ground Truth
VLMPlanner	Vision-Language Navigation planner module
Mst	Short-term memory
Mlt	Long-term memory
F	Frontier set
R	Region set
O	Object set
P	Agent position / path coordinate

1. Introduction and Background

1.1 Research Motivation

Vision-Language Navigation guided by high-level natural language instructions has emerged as a crucial task in the field of embodied artificial intelligence and robotics. The ultimate goal of Vision-Language Navigation is to develop autonomous agents capable of interpreting complex human commands and executing them safely within dynamic, unknown environments. However, current benchmarks and methodologies predominantly focus on short-horizon, single-stage tasks, such as instructing a robot to "go to the kitchen" or "find the apple". While these reactive policies have proven effective in highly controlled or simplified settings, they are fundamentally insufficient for real-world applications.

In realistic domestic or industrial scenarios, human instructions inherently encompass multiple sub-goals and demand long-horizon navigation. A practical robotic assistant must be able to execute continuous, long-sequence tasks, such as finding the towel in the bathroom, taking it to the kitchen island, and finally bringing the cup to the living room. Executing such tasks requires a paradigm shift from simple target-driven movement to sustained spatial reasoning, cross-task learning, and adaptive re-planning over extended periods. As the field transitions toward these Long-Horizon Vision-Language Navigation challenges, the demand for robust, long-term memory architectures has become the most pressing bottleneck.

Furthermore, in real-world deployments, robots face the ultimate test of zero-shot navigation, operating in entirely unseen environments without prior mapping. This highlights the urgent need for navigation systems that rely on explicit, real-time spatial reasoning and dynamic experiential learning, enabling them to adaptively construct knowledge and make informed decisions on the fly.

1.2 Problem Definition

The core challenge in Long-Horizon Vision-Language Navigation lies in the limited memory capacity of the agent and its inability to reason across sequential tasks in strictly zero-shot scenarios. When navigating complex and unknown environments, current robotic systems face five critical limitations:

1.2.1. The Semantic-Spatial Disconnect in Decision Making

Existing models struggle to seamlessly bridge high-level semantic reasoning with precise physical execution. An agent might utilize a Large Language Model to infer the common-sense knowledge that a facial

cleanser is typically located near a bathroom sink, or it might retrieve this fact from an independent experiential memory module based on past observations. However, this semantic prior remains unactionable without a high-fidelity spatial representation that provides structural relationships. Traditional two-dimensional representations, and even unstructured three-dimensional coordinate systems, lack fine-grained geometric details and topological object dependencies. Consequently, the semantic targets generated by reasoning models lack strict spatial grounding. This disconnect prevents the agent from effectively translating abstract knowledge or historical experiences into concrete, navigable actions, leaving it unable to formulate a coherent execution strategy even when the theoretical location of a target is known.

1.2.2. Lack of Cross-Task Reasoning and Knowledge Transfer

Throughout the execution of multi-stage instructions, agents continuously generate valuable experiential data, including the confirmation of inaccessible rooms and the deduction of object co-occurrence patterns. However, current systems lack the capacity to extract these successful search heuristics or failure traps from completed sub-tasks. Without robust cross-task knowledge transfer, agents are unable to apply historical context to future actions, leading to the repetition of exploration errors and the execution of redundant navigational decisions.

1.2.3. Unrealistic Prior Reliance and Blind Exploration

Many active exploration algorithms implicitly rely on pre-built offline representations, including static topological maps, two-dimensional occupancy grids, and pre-computed boundary maps. This dependency restricts their online operations to mere updates of these prior structures, acting as an unrealistic shortcut that strictly violates the principles of zero-shot navigation in completely unknown environments. Furthermore, when navigating without these structural priors, traditional methods typically select exploration frontiers based solely on physical distance. This purely distance-driven approach lacks global semantic guidance, causing the agent to inefficiently wander in low-value areas within large-scale scenes.

1.2.4. Memory Overload and Cognitive Hallucinations

Continuous exploration over extended horizons generates an exponential accumulation of sensory and semantic data. Retaining all historical high-resolution observations to maintain perceptual accuracy rapidly exhausts the local computational memory of the robot and induces severe latency during interactions with foundational models. Conversely, aggressively compressing these memories eliminates critical environmental context. This loss of detail invariably leads to cognitive hallucinations, where the reasoning model conflates current spatial states with historical ones. Without an active and strategic forgetting mechanism, systems inevitably

suffer from memory explosion, computational overload, and degraded reasoning performance.

1.2.5. Partial Observability and Compounding Execution Errors

In expansive environments, agents operate under strict partial observability, forcing them to make sequential decisions based on incomplete information. In LH-VLN, a minor deviation or misjudgment in an early sub-task (e.g., failing to enter the correct room) can cause a complete breakdown of all subsequent sub-tasks. Current models often lack the error tolerance required to efficiently backtrack, recover, and re-plan when encountering a dead end, leaving them vulnerable to catastrophic error propagation across long task horizons.

1.3 Significance of the Work

This project aims to directly address the critical gap between spatial representation, physical execution, and experiential learning in long-horizon navigation. To overcome the limitations of existing methods, this research proposes a novel 3D Scene Graph Enhanced Memory System characterized by an integrated dual-memory architecture and an active exploration strategy.

Conceptually, this architecture mimics biological cognitive structures by dividing memory into two distinct channels. A hierarchical 3D semantic scene graph acts as the Hippocampus of the system, strictly maintaining the real-time geometric and semantic state of the physical world. Concurrently, an independent, vectorized experience module functions as the Cerebral Cortex, storing abstracted search strategies, past failures, and reasoning histories of the language model.

To guarantee efficient zero-shot navigation, this work also introduces a 4D Frontier Semantic Scoring Mechanism. This mechanism completely eliminates the reliance on ground-truth maps by actively evaluating unknown boundaries across four dynamic dimensions: Information Gain, Room Unexplored Ratio, Semantic Similarity, and Physical Distance.

The significance of this work lies in its ability to unify precise geometric understanding with task-history reasoning. By prioritizing exact spatial queries within the 3D scene graph and utilizing experiential memory as a fallback strategy for intelligent error recovery, the system heavily mitigates cognitive hallucinations and limits cross-stage error propagation. Simultaneously, memory explosion is prevented through strategic, contribution-based pruning. Ultimately, this research provides a scalable, robust, and highly generalizable framework for autonomous robotic assistants executing lifelong tasks in complex real-world settings.

2. Aims and Objectives

2.1 Core Aim

The core aim of this research is to design and implement a robust navigation system that integrates dynamic 3D scene graphs with a vectorized experiential memory to address the challenges of Long-Horizon Vision-Language Navigation (LH-VLN) in zero-shot, unknown environments. Operating entirely without prior maps, the system aims to eliminate blind exploration, prevent memory explosion, and facilitate cross-task knowledge reuse to achieve highly efficient and continuous navigation.

2.2 Specific Objectives

To fulfill the primary research aim, this project focuses on the following five specific objectives:

Hierarchical 3D Spatial Representation: To construct a dynamic, hierarchical 3D scene graph that manages building, room, and object topological relationships in real time, while embedding distinct exploration degree attributes within each room layer to provide structured perception for the central planner.

Cross-Task Experiential Memory Engine: To develop an independent, vectorized experience module capable of automatically extracting successful trajectories and failure traps, including hazardous area interception records, to facilitate knowledge reuse across sub-tasks while employing active pruning to prevent memory overload.

Zero-Shot Active Exploration Scoring Mechanism: To implement a four-dimensional scoring framework operating independently of ground-truth maps, which synthesizes information gain, room unexplored ratios, target semantic similarities, and physical distances to autonomously deduce optimal exploration directions in unknown spaces.

Physical Safety Assurance Protocol: To design a local geometric validation mechanism based on navigation meshes that incorporates coordinate snapping and a thirty-centimeter safety pull-back policy, thereby correcting zero-shot pathfinding errors and eliminating the risk of camera clipping.

Benchmark Evaluation and Ablation Studies: To validate the complete system on the Habitat simulation platform using the LHPR-VLN dataset across fine-grained metrics including the Independent Success Rate (ISR) and the Conditional Success Rate (CSR), followed by comprehensive

ablation studies to isolate the algorithmic contribution of each individual module.

3. Project Planning

3.1 Overall Architecture

The proposed navigation framework follows a perception-cognition-action paradigm, structured across three core dimensions:

Perception and Mapping (Vision): Operating without prior blueprints, the system reproduces the GraphEQA framework and integrates Hydra to perform real-time RGB-D segmentation. It constructs a dynamic, hierarchical 3D semantic scene graph to establish spatial awareness. Concurrently, an independent Vectorized Experience Memory is built to retain cross-task knowledge.

Cognitive Reasoning (Language): Serving as the decision engine, a Large Language Model utilizes a dual-channel retrieval mechanism. It dynamically evaluates task progress by prioritizing explicit spatial queries from the scene graph and retrieving historical heuristics from the experience memory, thereby avoiding redundant decisions.

Exploration and Execution (Navi): Semantic intentions are translated into physical kinematics. The system evaluates exploration frontiers using a 4D scoring mechanism and applies strict physical safety filters to guarantee collision-free trajectories within valid navigable zones.

3.2 Timeline and Execution

The project is executed over a rigorous six-month timeline:

Oct – Nov 2025: Conducted literature reviews, defined the architectural framework, and submitted the project planning and risk reports.

Nov 2025 – Jan 2026: Configured the Habitat 3D simulator and integrated Hydra 3D mapping. The original single-step task was reconstructed into a multi-stage Long-Horizon navigation pipeline. Successfully generated initial demonstrations and drafted the interim progress report.

Feb – Mar 2026: Addressed critical challenges identified during the mid-term evaluation. This included resolving Open3D GLFW rendering errors for 3D visualization, mitigating the over-reliance on GPT-generated confidence scores by designing internal exploration degree metrics, and handling dataset ambiguities regarding target object coordinates.

Mar – Mid-May 2026: Finalized core algorithmic modules. Developed the Vectorized Experience Memory, integrated active exploration degrees into the 3D scene graph, and overhauled navigation success logic. Improved 2D and 3D visualization tools, and conducted comprehensive system

debugging to resolve trajectory generation failures and environment interaction anomalies.

Late May 2026: Executed automated batch testing, calculated final evaluation metrics, and completed the dissertation and academic poster preparation.

3.3 Risk Assessment and Mitigation

To ensure structural robustness, several technical and deployment risks were preemptively mitigated:

Environment and Version Conflicts: The embodied AI toolchain involves numerous deep-learning libraries prone to version conflicts. This is mitigated by employing Anaconda to create strictly isolated virtual environments, ensuring dependency stability.

Headless Visualization Bottlenecks: Real-time graphical visualization of the Habitat simulator is severely impeded on headless remote GPU servers. This is mitigated by executing simulations entirely in a dedicated headless rendering mode and deploying automated scripts to capture sequential frames, which are then synthesized into video files for comprehensive post-execution trajectory analysis.

Memory Overload: Continuous exploration risks hardware exhaustion. This is resolved by converting high-resolution observations into lightweight vectorized embeddings and implementing a contribution-based pruning strategy.

Algorithmic Collisions: Reconstructed virtual environments contain geometric defects. The execution layer prevents system crashes by enforcing a strict navigable mesh validation check paired with a physical safety pull-back mechanism.

4. Literature Review

4.1 The Paradigm Shift in Navigation Architectures

The fundamental objective of Vision-Language Navigation (VLN) is to establish a robust mapping from multimodal inputs, which consist of natural language instructions and egocentric sensory observations, to a sequence of navigable actions [1]–[3]. Historically, the architectural design of these navigation systems has evolved through several distinct milestones, shifting from monolithic end-to-end networks to highly decoupled modular systems.

4.1.1 End-to-End Paradigms and Their Generalization Bottlenecks

Early approaches to VLN predominantly adopted single-system end-to-end paradigms. These frameworks utilized monolithic neural networks, such as those trained via Decentralized Distributed Proximal Policy Optimization (DD-PPO), to map raw visual features directly to low-level motor commands. Recently, researchers have extended this paradigm by employing large Vision-Language-Action (VLA) foundation models to implicitly align multimodal features for reactive obstacle avoidance.

While these advanced end-to-end architectures exhibit impressive interactive fluidity, they possess critical vulnerabilities in Long-Horizon Vision-Language Navigation (LH-VLN) tasks [4]–[7]. The primary bottleneck lies in their inability to manage persistent semantic memory. Because these systems rely on implicit neural weights rather than explicit geometric structures, they cannot reliably anchor past experiences to concrete physical coordinates. This structural deficiency inevitably leads to severe compounding execution errors and cognitive hallucinations when agents operate under strict zero-shot conditions.

4.1.2 The Transition to Modular and Dual-System Architectures

To bypass the generalization bottlenecks inherent in end-to-end models, recent research has pivoted toward modular paradigms and dual-system architectures. Frameworks such as DualVLN explicitly decouple the navigation pipeline into specialized, heterogeneous sub-components representing a slow cognitive system for high-level semantic planning and a fast reactive system for low-level geometric locomotion [8],[10],[11],[12],[15].

This modular decoupling provides a crucial advantage because it allows for the integration of explicit, persistent memory structures. By separating the cognitive planner from the low-level controller, modular systems empower the agent to maintain stable environmental records and execute autonomous backtracking when high-level plans fail. However, these

systems still depend primarily on implicit neural representations rather than persistent explicit spatial memory. They remain vulnerable to spatial drift, historical inconsistency, and cognitive hallucination in zero-shot long-horizon environments. Consequently, the central research challenge has shifted toward architectural design, specifically focusing on how an autonomous agent should store, retrieve, and manage spatial and experiential memory over extended operational horizons.

4.2 Map-Based Spatial Storage

Within modular frameworks, the design of long-term spatial memory plays a central role in determining the agent’s ability to ground language in space and support extended exploration. Existing map-based storage approaches can be broadly divided into two categories: 2D semantic maps and topological graphs.

4.2.1 2D Semantic Maps

Traditional modular systems, exemplified by frameworks such as MoViLan, rely on 2D semantic grid maps as their core spatial memory [6],[18],[19],[20]. These methods project pixel-level semantic predictions from egocentric cameras onto a static top-down occupancy plane. Similarly, Wani et al. introduced the MultiON benchmark, where their Semantic Map baseline demonstrates that explicit spatial memory significantly outperforms implicit recurrent neural network memory. Furthermore, MEM2EGO attempts to bridge global environmental memory and local perception by converting these maps into textual descriptions to guide VLM-based navigation.

Although 2D semantic grids are effective for capturing horizontal obstacle boundaries, they suffer from substantial geometric limitations. By collapsing the environment into a single plane, these representations discard the vertical axis and therefore fail to distinguish spatially layered structures, such as items placed on different shelves or objects positioned underneath furniture. As a result, reasoning modules lose access to crucial 3D physical context, restricting the precision of navigation decisions.

4.2.2 Topological Maps

To improve computational scalability and reduce memory cost, other works have explored sparse topological graphs. Models such as MG-VLN and VLN-KHVR abstract the environment into discrete nodes representing viewpoints or room-level locations, with edges encoding navigable connectivity [3],[11],[15],[21],[22].

While these topological representations are well suited to macro-level routing over long distances, their abstraction is often too coarse for precise target localization. Because they omit continuous 3D metric

boundaries, sparse graphs cannot provide sufficiently dense geometric constraints when the agent transitions from room-level traversal to local physical execution. This limitation often leads to stalled movement or invalid trajectories near the goal.

4.3 Independent Memory Modules

In parallel with spatial mapping research, another branch of work focuses on independent memory modules that are decoupled from physical geometry and instead aim to capture historical experience, sequential decisions, and task-level heuristics.

4.3.1 Vectorized Storage

In experience-driven navigation, systems compress historical execution data into dense mathematical representations [23],[24],[25]. Anwar et al. developed ReMEmbR, which is a retrieval-augmented memory architecture that encodes a robot’s historical deployment data, including observations and timestamps, into high-dimensional dense vectors to support long-horizon contextual recall [29]. Similarly, StreamVLN adopts a streaming memory strategy in which long-term histories are compressed using 3D-aware token pruning to maintain a manageable context size[30]. Additionally, the LH-VLN benchmark introduces demand-driven navigation baselines that combine transient mapping with vectorized sub-goal generation to support abstract long-horizon planning.

4.3.2 Direct Token Storage

Rather than compressing history into vector embeddings, some approaches directly store sequential textual records within the context window of Large Language Models. CLOI-NAV represents a prominent example of this paradigm, where ordered execution logs are appended into the prompt context to preserve temporal dependencies and multi-step instruction states [25]. This text-centric strategy enables agents to leverage in-context learning for trial-and-error reasoning without requiring additional learned memory modules. However, this unprocessed storage method is prone to causing memory explosions.

4.3.3 The Semantic-Spatial Disconnect

Despite their advantages in reasoning, purely experience-centric systems suffer from a fundamental semantic-spatial disconnect. The stored information remains detached from any continuous 3D physical coordinate system, which means that historical knowledge is preserved as ungrounded data rather than as spatially actionable memory. An LLM planner may correctly infer from prior logs that a room was previously a dead end, but without geometric grounding, such knowledge is often misapplied. The agent may confuse the present position with a structurally

similar historical node, resulting in cognitive hallucinations where the system proposes logically sound but spatially infeasible navigation goals.

4.4 Emerging Hybrid Architectures and Agentic Frameworks

To bridge the gap between abstract semantic experience and dense metric geometry, recent embodied AI research has increasingly explored hybrid architectures that combine spatial memory, streaming context management, and agentic decision-making.

4.4.1 Hybrid Architectures and World Models

Cutting-edge frameworks such as Dynam3D and WMNav integrate Vision-Language Models with predictive world modeling [26],[27]. These systems allow the agent to simulate future visual rollouts in an imagined space and evaluate multiple trajectory candidates before executing physical actions. While generative planning reduces real-world collision rates through prospective evaluation, these approaches require immense computational resources and are highly susceptible to accumulated imagination errors.

4.4.2 Agentic Navigation Frameworks

Another emerging direction positions Vision-Language Models as active agentic controllers. Frameworks such as AgentVLN and RoboClaw utilize agentic reasoning loops in which VLMs dynamically orchestrate modular perception, planning, and execution skills [28],[29]. These systems demonstrate strong autonomous replanning and error-correction capabilities. However, most existing agentic frameworks still rely primarily on 2D visual prompts or sparse topological abstractions, which maintains a representational gap between the high-level cognitive brain and the dense 3D physical bounds of the environment.

4.5 3D Semantic Scene Graphs as a Cognitive Substrate

To achieve a lossless synthesis of semantic hierarchy and metric precision, recent works have begun deploying 3D semantic scene graphs for embodied navigation. Frameworks such as SayPlan and ConceptGraph demonstrate that scene graphs can organize physical space into an explicit, symbolic network layered across building, room, and object levels [30],[31],[32]. Each object node retains its exact 3D metric coordinates and oriented bounding boxes while explicitly storing topological dependencies.

Building upon this structural paradigm, GraphEQA serves as a direct architectural inspiration for this dissertation [32]. The multi-layered representation of 3D scene graphs provides LLM planners with a highly

compressed yet geometrically rigorous environmental abstraction, establishing it as the ideal deterministic cognitive substrate for anchoring abstract historical experiences onto concrete physical entities. However, most existing scene-graph-based navigation systems treat the graph merely as a static environmental representation and lack independent experiential channels for recording cross-task trial-and-error heuristics.

4.6 Identified Research Gaps

A comprehensive review of the literature reveals three major gaps in zero-shot long-horizon vision-language navigation that directly motivate this dissertation:

First, there is an absence of dual-channel spatial-experiential integration. Existing methods typically focus either on objective spatial mapping or on subjective historical memory, but rarely fuse the two in a way that is architecturally explicit. Current systems lack a parallel strategy mechanism that can preserve trial-and-error heuristics while maintaining a grounded 3D representation of the environment.

Second, current memory management strategies are inadequate for lifelong or extended-horizon exploration. Many methods rely on additive memory growth, which leads to increasing computational burden and reasoning latency. Existing compression heuristics, such as simple temporal truncation, remove information too uniformly and fail to account for the relative contribution, task relevance, and spatial reachability of different memory elements.

Third, exploration evaluation still depends heavily on privileged information. Many active exploration methods assume access to ground-truth maps or similar oracle-level data when computing room coverage or information gain. In realistic zero-shot settings, such assumptions are impractical. There remains a lack of formal exploration frameworks that can estimate local completion and semantic potential purely from frontier visibility and scene graph extrapolation.

5. Methodology

The core engineering contribution of this thesis is the design, mathematical formulation, and implementation of an online, multi-modal memory architecture integrated with a hierarchical planning paradigm for Long-Horizon Vision-Language Navigation (LH-VLN) in zero-shot, unmapped environments. Inspired by the architectural principles of GraphEQA, which combines global, semantically-sparse spatial structures with local, semantically-rich visual contexts, this framework addresses the three primary bottlenecks of long-horizon tasks: blind exploration, exponential memory growth, and the disconnect between high-level language planning and low-level physical locomotion.

The system decomposes the navigation pipeline into three interconnected layers: perception and mapping (Vision), cognitive reasoning and retrieval (Language), and path planning and safety execution (Navi).

5.1 Hierarchical 3D Spatial Memory Construction (The "Hippocampus")

The foundational sub-system functions analogously to a biological hippocampus, incrementally building and maintaining a real-time structural model of the physical world without any pre-existing environment blueprints or global maps. Spatial perception operates through continuous ego-centric RGB-D sensor streams and spatial pose inputs.

5.1.1 Volumetric TSDF Fusion and Metric Mapping

As the robotic agent navigates the scene, dense three-dimensional geometric structures are recovered online using a Volumetric Truncated Signed Distance Function (TSDF) fusion pipeline. Incoming depth frames are projected into a discrete three-dimensional voxel grid based on camera intrinsics and real-time odometry transforms. Each voxel integrates depth observations to recursively update its distance values, classifying spatial regions into free, occupied, or unmapped voxels.

Concurrently, a semantic segmentation stream (utilizing open-set models such as Detic in physical deployments or ground-truth semantic IDs in simulated benchmarks) extracts pixel-level instance masks from the corresponding RGB frames [37]. These 2D semantic masks are back-projected into the 3D voxel space, utilizing probability fusion to aggregate semantic category labels over sequential frames, creating a dense, high-fidelity metric-semantic 3D voxel representation.

5.1.2 Topographical Scene Graph Generation via Hydra

To transform this dense, computationally expensive metric grid into a compact symbolic network suitable for high-level language planning, the system reproduces the Hydra architecture to construct an incremental directed topological network graph utilizing a graph-theoretic framework [34]. The environment is abstracted into a multi-tiered graph structure comprising five distinct vertical layers:

Layer 5 (Top): The global building node unifying all lower architectural and topological spaces.

Layer 4: Functional room nodes representing enclosed structural partitions of the building layout.

Layer 3: Explored region or place nodes summarizing localized navigable spaces.

Layer 2: Object nodes containing clustered instance labels, bounding boxes, and centroid positions, alongside agent nodes tracking the current position of the robot.

Layer 1 (Base): The dense metric-semantic 3D mesh representing raw surface geometry.

Intra-layer edges denote traversability boundaries, whereas inter-layer edges denote structural belonging properties.

Furthermore, to compress structural descriptions for the Large Language Model planner, the system implements an active topology compaction mechanism. When region tracking is disabled, the system dynamically purges all layer-three region nodes from the active graph topology. It automatically re-routes structural edges to connect room nodes directly to child objects and agents, constructing direct room-to-object and room-to-agent pathways. This adaptive graph collapse effectively reduces prompt token lengths and simplifies the context space for the cognitive layer.

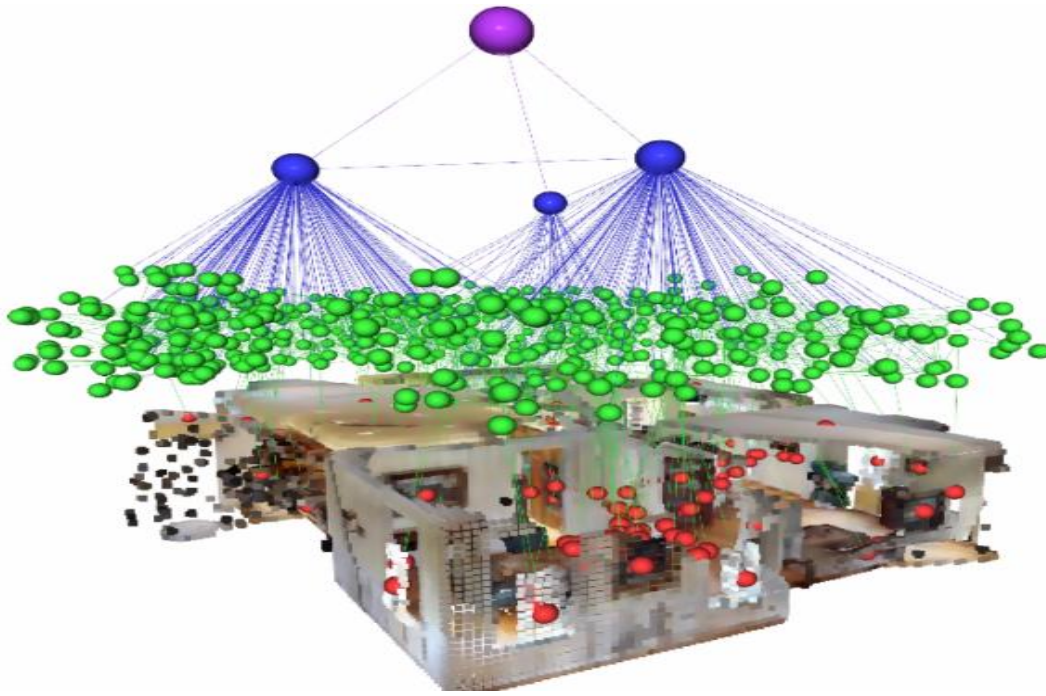


Figure 5.1 Visualization of 3D topological scene graph

Semantic Topology (Room & Objects)

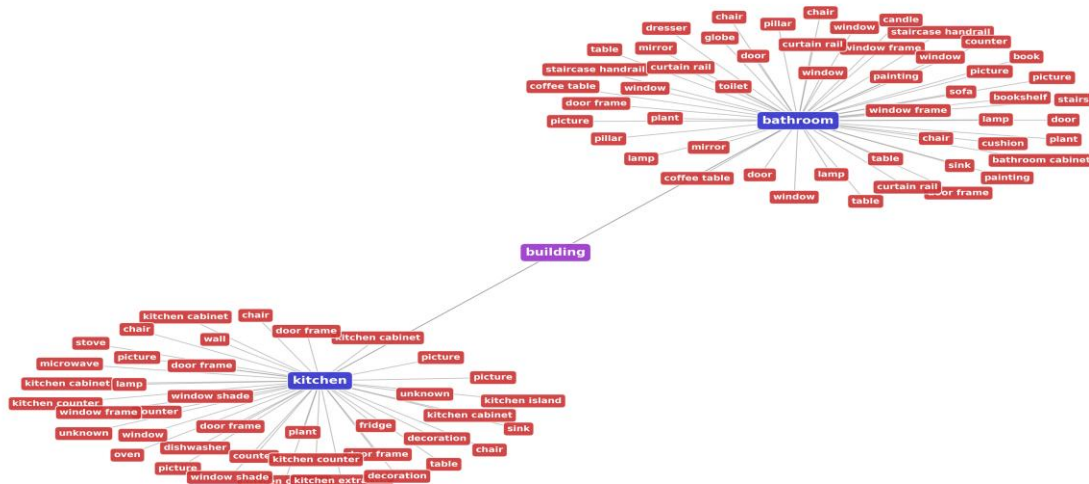


Figure 5.2 Visualization of 3D topological scene graph

5.1.3 Region-Level Exploration Metrics and Room Labeling

To ground exploration heuristics, this architecture introduces a localized coverage evaluation metric embedded directly within the layer-three region nodes. Each region node maintains an *exploration degree* parameter that continuously quantifies the volume of unknown frontier voxels and active boundary state flags derived from the underlying mapping grid.

During active exploration, newly generated candidate frontiers are structurally assigned to an existing region node via a region-to-frontier edge if their linear distance satisfies a close-range proximity threshold of less than 5m. This spatial pairing triggers the accumulation of a constant exploration weight parameter onto the region's overall score, constructing a real-time exploration confidence topology that supplies quantified room-level remaining exploration values to the high-level planner.

Concurrently, a language-driven semantic enrichment pipeline resolves the generic sequential identifiers natively assigned to layer-four room spaces. When a room vertex clusters a stable group of child object vertices, the unique object labels within that sector are harvested and processed via an isolated, structured Large Language Model prompt:

"Given the list of objects: [object_names]. Which room are these objects most likely found in? Keep explanation very brief."

To guarantee strict programmatic type safety during this open-world inference, the text response is bound to a structured schema enforced by a human-centric room type enumeration class. The resolved semantic label overwrites the functional name attribute of the layer-four room vertex in real time, transforming abstract geometric enclosures into descriptive semantic maps optimized for context-grounded reasoning.

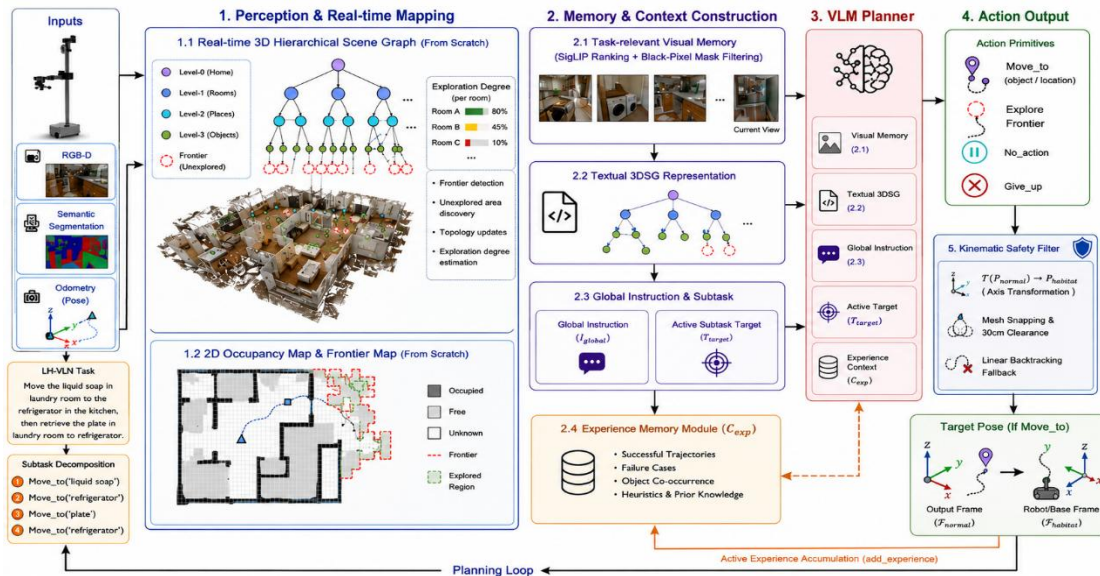


Figure 5.2 Overall architecture of the Dual-Channel Memory framework

5.2 Vectorized Experience Memory Engine

To transcend the limitations of traditional spatial maps that lack historical causal reasoning and temporal logic, the framework incorporates an online Vectorized Experience Memory module. Operating parallel to the short-term hierarchical spatial representation, this module functions as an

abstract cognitive engine designed to retain, retrieve, and consolidate multi-stage decision heuristics across sequential sub-tasks.

5.2.1 System Architecture and Node Formalization

The experience memory engine is structured as an agile, high-dimensional vector database composed of discrete historical nodes [23],[24]. To ensure perfect alignment with the programmatic pipeline, each experience node is formalized as a compact multi-factor tuple containing only the variables optimized within the mathematical vector space:

$$E_i = (V_i, o_i, \gamma_i)$$

Where the constituent elements are defined as follows:

- $V_i \in \mathbb{R}^{1536}$ represents the dense semantic text embedding vector generated via a contrastive embedding model, capturing the formalized historical query string.
- $o_i \in \{\text{Success, Failure, Exploration}\}$ denotes the definitive physical execution outcome of the logged navigation phase.
- γ_i represents the spatial anchor identifier mapping the memory node directly to a specific coordinate vertex within the hierarchical 3D scene graph.

By anchoring abstract semantic vector profiles directly to explicit geometric landmarks inside the topological map, the architecture avoids the memory exhaustion associated with continuous raw image buffer storage while equipping the agent with cross-task knowledge retention.

5.2.2 Symmetric Semantic Encoding and Dual-Objective Retrieval

At any discrete planning interval, the experience engine guides context-grounded reasoning through a symmetric, dual-objective retrieval channel. To minimize high-dimensional alignment errors caused by matching variable-length runtime text inputs against diverse text block matrices, incoming sub-task commands are pre-processed using a structural symmetry protocol:

$$V_q = \psi(q_{\text{current}})$$

where ψ represents the text embedding transformation that maps the normalized runtime query template into the identical 1536-dimensional vector space utilized by the memory ledger.

When a new search sequence is initialized, the module evaluates the entire active database against both semantic relevance and physical execution constraints [23],[24],[25]. For each stored experience node E_i , a composite retrieval value index is dynamically derived:

$$\text{Score}_{\text{retrieval}}(E_i) = w_{\text{rel}} \cdot \left(\frac{V_q \cdot V_i}{\|V_q\| \|V_i\|} \right) + w_{\text{reach}} \cdot \max \left(0, 1 - \frac{\text{Dist}_{\text{geo}}(P_{\text{agent}}, P_i)}{D_{\text{max}}} \right) + \Omega(E_i)$$

In this formulation, the mathematical components are structured as follows:

- The first term leverages a high-dimensional dot product to compute the precise cosine similarity between the current active command and the archived context vector, applying an empirical filtering threshold (0.3) to eliminate irrelevant background noise.
- The second term computes the spatial reachability coefficient, where $\text{Dist}_{\text{geo}}(P_{\text{agent}}, P_i)$ measures the shortest physical distance across the navigable mesh from the agent's current position P_{agent} to the memory's structural anchor coordinate P_i , normalized against a localized maximum bounding perimeter D_{max} .
- The parameters w_{rel} and w_{reach} serve as positive scaling coefficients balancing semantic priority against proximity overhead.

To safeguard the agent against repetitive kinematic deadlocks in complex multi-room layouts, a discontinuous pitfall mitigation modifier $\Omega(E_i)$ is integrated into the sorting matrix:

$$\Omega(E_i) = \begin{cases} 0.4, & \text{if } o_i = \text{Failure} \wedge \text{Dist}_{\text{geo}}(P_{\text{agent}}, P_i) \leq 2.5 \text{ m} \\ 0.0, & \text{otherwise} \end{cases}$$

If the positioning sensors detect that the robot is physically encroaching within a 2.5-meter radius of a coordinate where a previous sub-task failed, the modifier injects a dominant priority bias into that node. This mechanism forces the planner to extract negative experience payloads during spatial crises, embedding historical roadblock warnings directly into the active vision-language prompt window to prevent looping errors.

5.2.3 Predictive Context-Aware Memory Pruning

To preserve computational bounds and maintain low processing latency over expansive exploration horizons, the memory engine deploys a strategic retention mechanism driven by a multi-factor value assessment script. When the total node count breaches the steady-state hardware storage capacity threshold, the system calculates a forward-looking contribution index $C(E_i)$ to perform active forgetting:

$$C(E_i) = \text{Relevance}(E_i, \{q_m\}_{m=k+1}^M) \times \text{Reachability}(P_{\text{agent}}, P_i)$$

This optimization protocol decouples database cleaning from simple chronological deletion or historical frequency counters, relying instead on

an anticipatory cognitive framework composed of two components [23],[24]:

- **Predictive Sub-Task Relevance:** The system extracts the array of remaining, unexecuted downstream sub-tasks $\{q_m\}_{m=k+1}^M$ derived from the global instruction string. It computes the maximum cosine similarity between the candidate memory embedding V_i and the semantic vectors of all upcoming tasks:

$$\text{Relevance}(E_i, \{q_m\}_{m=k+1}^M) = \max_{m \in [k+1, M]} \left(\frac{V_i \cdot \psi(q_m)}{\|V_i\| \|\psi(q_m)\|} \right)$$

This proactive look-ahead check protects contextual templates required for upcoming reasoning stages from premature eviction, ensuring that critical knowledge lines are preserved before the robot physically reaches those execution steps.

- **Spatial Reachability:** The physical proximity component measures the localized accessibility of the memory's geometric anchor node relative to the robot's coordinates, scaling down the retention value of experiences locked inside distant, sealed, or completely isolated topological chambers.

The memory manager executes this dynamic value assessment across the entire active list, identifies the absolute lowest-scoring node as non-contributing redundant data, and systemically ejects it from the memory matrix. This active forgetting protocol bounds memory overhead to a steady-state threshold without sacrificing semantic precision or grounding performance, resolving the data-explosion bottlenecks inherent to extended long-horizon robotic deployment.

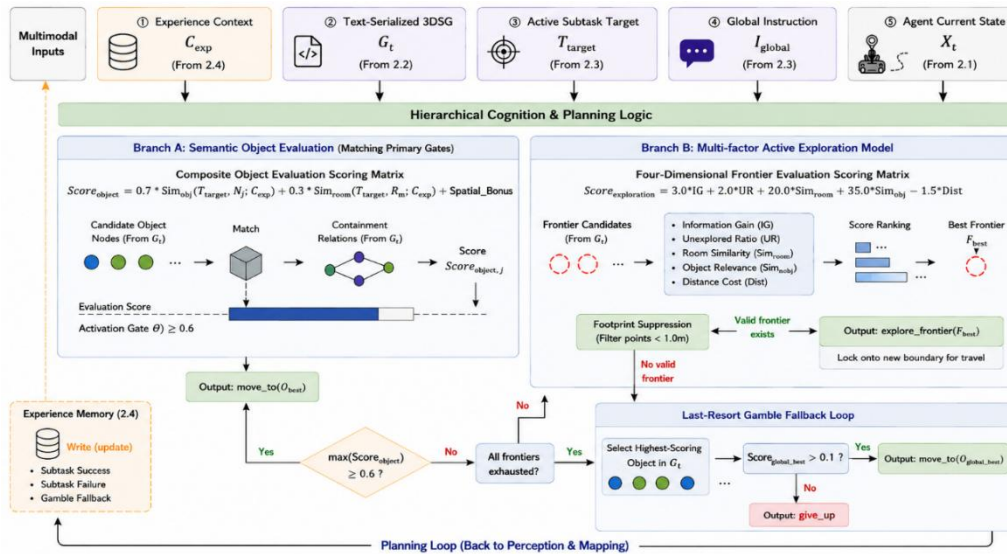


Figure 5 System Architecture and decision logic for VLM planner

5.3 Vision-Language Navigation Decision Layer

The vision-language navigation decision layer acts as the primary cognitive decision-making hub of the navigation framework. Implemented online via the `VLMPlannerLHPRVLNGPT` planner class, this module handles long-horizon vision-language navigation (LH-VLN) tasks by combining high-level semantic reasoning with incremental 3D scene graphs and experience memory banks. The planner decomposes complex global instructions into discrete sequential sub-tasks, mapping environmental attributes against topological nodes to output reliable navigation targets.

5.3.1 Inputs and Multi-Modal State Formalization

At each discrete decision interval t , the planning architecture constructs a multi-modal state payload. To enable the core language model to process grounding information without parsing heavy spatial data, the system flattens the structural scene graph into a compact, text-serialized topology tracking visible object nodes high-level semantic reasoning and planning in our framework are driven by the gpt-4o-mini model provide, registered room definitions, and unvisited frontiers. The complete input state configuration is formalized as a structured multi-modal tuple:

$$S_t = (I_{\text{global}}, s_{\text{current}}, G_t, X_t, C_{\text{exp}})$$

Where the constituent variables are defined as follows:

- I_{global} represents the global natural language instruction string defining the overarching user macro-command (`vlm_instruction`).
- s_{current} represents the active sub-task dictionary tracking the target text label T_{target} .
- G_t represents the active directed topological scene graph managed via NetworkX, encapsulating extracted object nodes, room nodes, and active frontiers.
- X_t defines the immediate state string of the agent, formatted as: "The agent is currently at node <node_id> at position [x, y, z] in room <room_id>".
- C_{exp} represents the historical experience context (`exp_context`) extracted from the memory database, sorted and injected into the prompt stream to act as a contextual anchor.

5.3.2 Composite Object Valuation Engine

For spatial navigation primitives (`move_to`), the decision layer switches to a probabilistic verification mode, evaluating every discovered item within the active topological scene graph against a composite semantic scoring matrix [32],[38]. For each candidate object node O_j associated with an architectural room name R_j , the system computes an objective matching coefficient:

$$\text{Score}_{\text{object}}(O_j) = 0.7 \cdot \text{Sim}_{\text{obj}}(T_{\text{target}}, N_j) + 0.3 \cdot \text{Sim}_{\text{room}}(R_{\text{target}}, R_j) + \Phi_{\text{spatial}}$$

The sub-metrics driving this valuation engine are derived as follows:

- $\text{Sim}_{\text{obj}}(T_{\text{target}}, N_j)$ tracks the high-dimensional cosine similarity score evaluated by the semantic analyzer matching the sub-task target text T_{target} against the open-vocabulary linguistic label N_j of the graph node, conditioned on C_{exp} .
- $\text{Sim}_{\text{room}}(R_{\text{target}}, R_j)$ captures the room-level spatial alignment. The system utilizes a language model to extract the target destination room type R_{target} from the global instruction, matching it against the candidate object's parent room node label R_j . If no explicit room context is specified, this sub-metric defaults to a neutral baseline coefficient of 0.5.
- Φ_{spatial} is a discrete containment bonus modifier introduced to reward explicit spatial dependencies or structural containment properties:

$$\Phi_{\text{spatial}} = 0.2 \quad \text{if SpatialRelation is True,} \quad \text{otherwise} \quad 0.0$$

The composite score $\text{Score}_{\text{object}}$ is strictly bounded within the range of 0.0,1.0. The decision layer enforces a strict confidence activation gate: only when the highest-scoring candidate satisfies $\max(\text{Score}_{\text{object}}) \geq 0.6$ does the planner lock the target destination, setting $\text{action_output} = \text{'move_to'}$ and switching $\text{is_confident} = \text{True}$ to drive the physical controller. If all candidates fail to satisfy this threshold, the layer rejects the candidate pool and triggers the active frontier exploration framework.

5.4 Active Frontier Exploration Framework

5.4.1 Four-Dimensional Frontier Valuation Matrix

When short-term spatial memory lacks high-confidence navigational targets, the agent transitions into an active exploration state. Candidate navigation pathways are continually gathered by harvesting discrete frontier nodes from the underlying metric map. To determine the most promising exploration vector, the framework executes a comprehensive four-dimensional scoring operation across all unvisited frontier vertices F_i [15],[23],[24]:

$$\begin{aligned} \text{Score}_{\text{exploration}}(F_i) = & \alpha \cdot \text{IG}(F_i) + \beta \cdot \text{UR}(R_m) + \gamma_{\text{room}} \cdot \text{Sim}(T_{\text{target}}, R_m) \\ & + \gamma_{\text{obj}} \cdot \max_{O \in \mathcal{N}(F_i)} \text{Sim}(T_{\text{target}}, O) - \delta \cdot \text{Dist}(P_{\text{agent}}, F_i) \end{aligned}$$

The physical and architectural terms driving this optimization formula are defined as follows:

- $IG(F_i)$ represents the **Information Gain** metric, which evaluates local obstacle clutter density within a 1.5-meter radius to estimate the volume of unknown space lying beyond the frontier boundary.
- $UR(R_m)$ represents the **Room Unexplored Ratio** extracted from the parent room vertex R_m containing the frontier. It models the remaining exploration value of that specific room partition based on unmapped frontier voxels:
- $UR(R_m) = \min\left(10.0, \frac{\text{Voxels}_{\text{frontier}}(R_m)}{50.0}\right)$
- $\text{Sim}(T_{\text{target}}, R_m)$ captures the **Macro-Spatial Semantic Potential**, measuring the semantic similarity between the sub-task target T_{target} and the functional room label R_m conditioned on the experience context C_{exp} .
- $\max_{O \in \mathcal{N}(F_i)} \text{Sim}(T_{\text{target}}, O)$ isolates the **Micro-Contextual Semantic Potential**, scanning all known object nodes situated within a localized 2.0-meter physical radius $\mathcal{N}(F_i)$ of the frontier to leverage common-sense asset co-occurrence relationships.
- $\text{Dist}(P_{\text{agent}}, F_i)$ represents the **Kinematic Geodesic Decay** term, applying a distance penalty based on the path length from the agent's coordinates P_{agent} to the frontier position to enforce movement efficiency.

The scaling weights are configured to favor immediate semantic relevance over close geometric distance: $\alpha = 3.0, \beta = 2.0, \gamma_{\text{room}} = 20.0, \gamma_{\text{obj}} = 35.0, \delta = 1.5$. The system ranks all active gateways based on this composite metric, selects the best_frontier , and commands the low-level base controller to advance.

5.4.2 Frontier Exhaustion and Fallback Strategy

To maintain stability across continuous task frames, the exploration layer tracks environmental coverage to prevent redundant loop tracking. Frontiers that fall within a 1.0-meter physical radius of the agent are automatically logged into a visited coordinate array to suppress re-sampling. If a sub-task target change is detected ($T_t \neq T_{t-1}$), this visited tracking array is instantly cleared to reset the exploration state machine for the newly assigned target.

When all unvisited frontiers are fully exhausted and no active graph object satisfies the primary navigation threshold ($\text{Score}_{\text{object}} \geq 0.6$), a fallback routing strategy is executed to prevent the navigation sequence from failing prematurely [25],[26],[27]. The planner scans the historical similarity cache accumulated since the beginning of the episode. If the highest recorded score clears a minimal baseline ($\text{Score}_{\text{global_max}} > 0.1$), the module overrides planning uncertainty by forcing $\text{is_confident} = \text{True}$. The system then commands the robot to move directly to the spatial centroid of this maximum-posteriority candidate ($O_{\text{global_best}}$). This strategy leverages

the open-vocabulary generalization of the similarity framework to tolerate naming variances and maximize the probability of landing within the task completion boundary, executing a definitive abortion command only when the spatial graph contains zero valid objects.

5.5 Kinematic Safety Filters and Reference Alignment

The physical locomotion layer serves as a safety guard for the navigation framework, translating high-level symbolic destinations into valid multi-point pathways while handling reference coordinate frame transformations.

To bridge the structural gap between the multi-modal planner and the simulation environment, the system executes a dual-coordinate transformation [34]. This mapping transposes the vertical altitude vector from the Cartesian planning frame ($\mathcal{F}_{\text{normal}}$, where Z is the vertical axis) into the native simulation environment frame ($\mathcal{F}_{\text{habitat}}$, where Y is the vertical axis):

$$\mathcal{T}(P_{\text{normal}}) \rightarrow P_{\text{habitat}}$$

To prevent the navigation process from aborting or terminating when a target is placed inside a non-navigable zone, such as inside a solid object volume, behind an unmapped structural barrier, or outside the multi-room navigation mesh—the layer enforces a kinematic safety filter. Before invoking the pathfinder, the system evaluates target navigability. If the position is invalid, an edge-snapping utility projects the coordinate onto the nearest standing plane of the navigation mesh. The controller then computes a directional unit vector from the agent to the snapped position, retracting the final destination exactly 30 centimeters back along this visual line-of-sight path to maintain collision clearance.

If this retracted coordinate remains completely non-navigable due to dense local obstacles, the controller triggers a linear backtracking routine to ensure path availability. The algorithm samples intermediate testing points spaced at regular 0.2-meter intervals along the straight line moving backward from the target toward the robot's current position. The pathfinder executes a shortest-path connectivity check for each step; the moment a valid path is verified, the loop breaks, and this closest accessible coordinate overrides the original target. This multi-stage safety pipeline guarantees that the agent advances to the absolute physical limit of the target zone rather than freezing or terminating the program sequence due to non-navigable boundary faults.

5.6 Multi-Modal Navigation Interaction Summary

To illustrate the high-frequency data streams and algorithmic coordination across the mapping, memory, decision, and execution modules, the complete system workflow is structured in Table 5.1.

Table 5.1: Multi-Modal Navigation System Workflow Matrix

Functional Layer	Input Data Channels	Core Algorithmic Components	Output Deliverables
Perception & Mapping	Egocentric RGB-D Streams, Instance Segments, Odometry	Volumetric TSDF Fusion, Hydra Topological Extraction, Linear Layer Collapsing	Hierarchical 3D Scene Graph JSON, Unexplored Voxel Volumes
Experience Database	Outcome Strings, Context Vectors, Graph Anchor IDs	Symmetric Structural Encoding, Cosine Similarity Weights, Proximity-Based Pruning	Heuristic History Payloads, Local Pitfall Warnings
Cognitive Decision Layer	Extracted Room Queries, Scene JSON, Experience Context	Composite Object Valuation, Four-Dimensional Frontier Evaluation, Last-Resort Gamble Logic	Target Action Primitives, Objective Destination Coordinates
Physical Locomotion	Action Primitives, Frontier Points, NavMesh Boundaries	Dual-Coordinate Transformation, Safety Boundary Retraction, Linear Backtracking	Raw Trajectory Evaluation Logs, Safe Base Velocity Commands

6. Results and Analysis

6.1 Experimental Settings

Datasets & Scene Assets: We evaluated our method on the newly proposed large-scale long-range vision-language navigation benchmark LHPR-VLN [16]. Each task consists of a sequence of sub-tasks that must be completed in order, and the agent is required to reason over both the current environment and the task history. To ensure diversity in 3D space and semantics, all our scene assets are derived from the HM3D dataset [35]. In the main experiments, we randomly selected 62 scene tasks containing complex multi-stage instructions for benchmarking [16],[35],[38]; in the ablation studies, we conducted rigorous controlled variable testing on 12 highly challenging representative tasks.

Evaluation Metrics: Following the LHPR-VLN benchmark, we evaluate our method using both conventional navigation metrics and fine-grained sub-task metrics. Standard metrics include SR (Success Rate), OSR (Oracle Success Rate), SPL (Success weighted by Path Length), and NE (Navigation Error). In the benchmark, SR, OSR, and SPL are computed only when all sub-tasks in a long-horizon instruction are completed correctly in order, while NE is measured at the final stop action.

In addition, we report the long-horizon metrics introduced by LHPR-VLN:

Independent Success Rate (ISR): Measures the success rate of each subtask being completed independently, without considering temporal dependencies between tasks. Its formula is:

$$ISR = \sum_{j=0}^M \sum_{i=0}^N \frac{S_{j,i}}{M \cdot N}$$

Conditional Success Rate (CSR): Measures the success rate of a sequence of tasks under the condition that the prerequisite tasks are met. It can very sensitively reflect the error accumulation and temporal dependencies in long-term tasks. Its formula is:

$$CSR = \sum_{j=0}^M \sum_{i=0}^N \frac{s_{j,i}(1 + (N - 1)s_{i-1})}{M \cdot N^2}$$

CSR weighted by Ground Truth (CGT) further weights CSR by ground-truth path difficulty:

$$CGT = \sum_{j=0}^M \sum_{i=0}^N \frac{P_i}{P} \cdot \frac{s_{j,i}(1 + (N - 1)s_{j,i-1})}{M \cdot N}$$

where $s_{j,i}$ denotes whether the i -th sub-task in task j is successful, M is the number of tasks, N is the number of sub-tasks in the task, P_i is the ground-truth path length of the i -th sub-task, and P is the total ground-truth path length of the task. ISR measures independent sub-task completion, CSR captures sequential dependency, and CGT further weights CSR by ground-truth path difficulty.

Baselines & Implementation Constraints: We compare our method with the representative baselines reported in the LHPR-VLN paper, including Random, InstructNav, GLM-4v prompt, NaviLLM (Pretrain), NaviLLM (Finetuned), GPT-4 + NaviLLM, and MGDM variants [16],[17],[33]. These baselines cover zero-shot prompting, finetuned navigation models, and LLM-assisted task decomposition. In contrast to the benchmark methods, which typically rely on multi-view RGB-D input, our framework is evaluated under a strict single front-facing view setting. This makes the task substantially harder, but it also better matches low-cost real-world deployment constraints.

6.2 Main Results and Comparison

Table 1. Performance comparison on 2–3 subtask tasks and 3–4 subtask tasks

Baseline numbers are taken from the LHPR-VLN benchmark; our row is from the current single-view evaluation setting.

Method	Type	2-3 subtasks					3-4 subtasks				
		SR	NE	ISR	CSR	CGT	SR	NE	ISR	CSR	CGT
Random	-	0	14.09	0	0	0	0	10.91	0	0	0
InstructNav	Zero-shot	0	10.80	0	0	0	0	8.38	0	0	0
GLM-4v prompt	Zero-shot	0	15.63	0	0	0	0	10.97	0	0	0
NaviLLM	Pretrain	0	12.11	0	0	0	0	10.04	0	0	0
NaviLLM	Finetuned	0	12.24	0	0	0	0	9.79	3.54	2.53	5.24
GPT-4 + NaviLLM	Pretrain	0	12.23	0	0	0	0	10	4.37	2.91	5.23
MGDM Two-stage	Finetuned	0	10.44	0	0	0	0	8.78	5.13	3.42	6.00
MGDM Alternate	Finetuned	0	3.54	0	0	0	0	1.23	4.69	3.30	5.83
Ours	-	1.92	5.29	40.14	28.74	26.88	1.82	5.12	42.29	28.31	26.51

Analysis:

Overall, our method achieved a high ISR of 42.41%, high CSR of 29.48%, high CGT of 28.03%, high OSR of 71.43%, SR of 3.17%, SPL of 1.88% and NE of 5.09m across all LH-VLN tasks, spanning 62 scenarios. Table 1 shows that our single-view system remains the most competitive even under a stricter sensing constraint. Compared with the benchmark baselines, our method achieves much stronger sub-task-level performance, especially in ISR, CSR, and CGT, while also maintaining a lower NE. This indicates that explicit 3D scene structure, memory-guided

retrieval, and frontier-based exploration help the agent preserve task consistency across sequential sub-goals. The 2–3 and 3–4 subtask settings also suggest that task length alone is not the only factor determining difficulty; repeated spatial structure and cross-stage contextual reuse can substantially affect execution quality.

Even though SR is still not high at present, our method has shown a significant improvement compared to other baselines in terms of navigating to a single object and using the information from previous navigation sub-tasks to navigate subsequent tasks.

6.3 Ablation Studies

Table 2. Ablation study on the main architectural components.

The ablation subset contains 12 highly challenging tasks [16],[17]. The table reports the effect of removing each major module from the full framework.

Architecture / Variant	SR (%)	ISR (%)	CSR (%)	OSR (%)	SPL (%)	NE (m)	Avg Steps
Baseline	0	22.22	11.60	90	0	5.69	73.50
w/o Scene Graph	0	23.53	8.72	88.89	0	5.17	72.11
w/o Memory	8.33	51.06	35.76	75.00	1.52	4.35	58.42
w/o Safety Filter	0	42.86	35.42	72.73	0	4.78	51.09
w/o Frontier Score	8.33	46.51	37.50	83.33	1.87	4.82	70.91
Ours (Full Framework)	8.33	51.16	38.89	75.00	1.52	4.08	56.75

Analysis:

The ablation results show that every module contributes to the final performance. Removing the scene graph weakens semantic grounding and lowers ISR and CSR, while removing memory reduces the agent’s ability to reuse previously discovered navigation experience, therefore causes a lower CSR. The safety filter is also important for preventing invalid target execution near non-navigable geometry. Finally, the frontier scoring module improves navigation stability by balancing exploration gain, semantic relevance, and distance. Overall, the full framework achieves the best trade-off between navigation accuracy and long-horizon robustness.

6.4 Detailed Dataset Statistics

Table 3. Detailed dataset statistics by subtask length.

The definitions of Task Steps, Nav Dis, Obj Num, Reentry Rate, and Area Association follow the LHPR-VLN benchmark.

Type	Task Steps	Nav Dis	Obj Num	Reentry Rate	Area Association
2 subtasks	61.81	10.33	108.38	25.00%	1.25
3 subtasks	52.88	8.82	275.50	93.18%	2.00
4 subtasks	46.39	7.76	260.64	100.00%	2.55

These statistics show that longer tasks tend to have shorter average step length and shorter average navigation distance per stage, while reentry rate and area association increase sharply. This suggests that later subtasks are more likely to occur in previously visited regions, which may reduce the effective difficulty of some long-horizon tasks.

6.5 Failure-Prone Target Categories

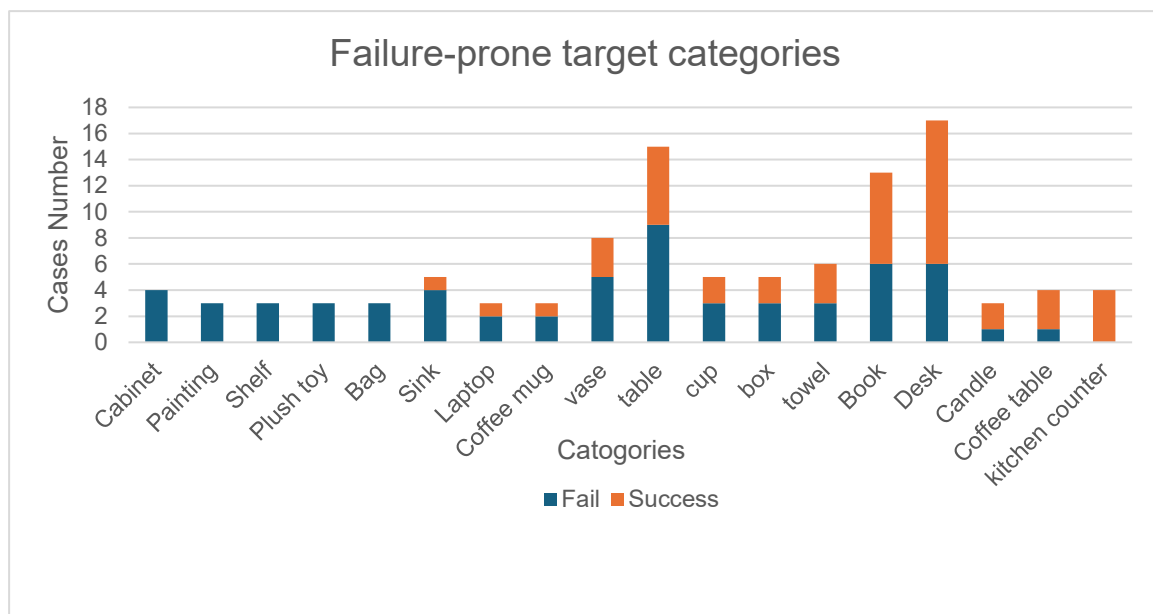


Figure 6.1 Failure-prone target categories (bar chart).

The most error-prone categories are cabinet, painting, shelf, plush toy, and bag, all of which reach a 100% failure rate in the current evaluation. Other difficult targets include sink, laptop, coffee mug, vase, table, cup, box, towel, book, and desk. In contrast, kitchen counter is the least problematic among the targets shown.

A bar chart is useful here because it makes the long-tail difficulty pattern immediately visible and highlights which object categories dominate the navigation errors.

7. Discussion

7.1 Main Results Analysis: Sequential Reasoning Under Single-View Constraints

The main experimental results show that the proposed framework substantially improves long-horizon navigation performance, especially at the sub-task level. Compared with the benchmark baselines in Table 1, the model achieves much stronger overall results across SR, ISR, CSR, CGT, OSR, and SPL, while also maintaining a lower navigation error. This suggests that the proposed architecture is effective not only at locating relevant semantic regions, but also at preserving task progress across sequential subtasks. The improvement is particularly notable given the strict single front-facing view constraint, which removes the lateral visual coverage commonly available in the benchmark baselines and makes local grounding significantly harder.

A key observation is the disparity between the relatively low SR and the much stronger ISR, CSR, and CGT. This is not contradictory. In LHPR-VLN, SR measures whether the agent completes the full sequence of navigation subtasks in the correct order, whereas ISR, CSR, and CGT measure sub-task completion and sequential consistency at a finer granularity. The results therefore indicate that the framework is already strong at maintaining partial progress and reasoning through intermediate goals, but that full end-to-end completion across all subtasks remains considerably more difficult under the benchmark’s strict success criterion.

The relatively high OSR provides a further clue. It shows that the agent often succeeds in reaching the correct semantic neighborhood, which confirms that the hierarchical 3D scene graph together with the VLM planner provides strong macro-level spatial grounding. However, the gap between OSR and SR also indicates that reaching the correct region is not always sufficient for full task success. Under the single-view constraint, the agent may correctly infer the destination area but still fail to maintain complete sequential consistency or to terminate the full long-horizon chain successfully. In that sense, the model demonstrates strong macro-navigation ability, but the remaining challenge lies in turning that regional success into complete task execution.

7.2 Ablation Analysis: Why the Full Framework Remains the Most Stable

The ablation study shows that the full framework achieves the most balanced overall behavior, even though some ablation variants may appear competitive on one or two isolated metrics. This is precisely why the results need to be interpreted jointly rather than one metric at a time. In particular, the ablation configurations often exhibit exploration or

navigation redundancy: they take more steps, wander more widely, and therefore gain more opportunities to accidentally enter the target object’s range. As a result, an ablated model can sometimes show a comparatively high OSR without actually being a stronger planner. This explains why a higher OSR does not necessarily imply better navigation quality, especially when it is accompanied by a lower SR or weaker sequential consistency.

The **w/o Scene Graph** condition illustrates this clearly. Once the 3D scene graph is removed, the agent falls back to a more geometry-driven exploration behavior. This can lead to a very high OSR, because blind or semi-blind exploration may eventually pass through the target neighborhood by chance. However, the same variant has SR and SPL of 0.00%, which shows that merely approaching the goal region is not sufficient when the system lacks semantic grounding. In other words, the agent may wander into the right area, but it cannot reliably recognize or complete the task in a semantically meaningful way. The contrast between high OSR and zero SR/SPL makes the difference between “accidental proximity” and “true task completion” very explicit. Furthermore, the absence of a scene graph means that the agent cannot recall locations it has visited or objects it has seen, which reduces the information reuse capability of the ablation method compared to the complete method.

The **w/o Frontier Score** condition provides a different kind of trade-off. In Table 2, this variant still achieves a reasonable SR, but it no longer benefits from the structured frontier valuation that balances information gain, semantic relevance, and distance. Even though the OSR of this variant is higher than that of the full framework in this 12-task subset, this does not imply a better policy. The average number of successful navigation steps is higher, and the SPL drops when the agent relies only on distance-based exploration. This means that the policy becomes less efficient: it may occasionally reach the target region, but it does so through more redundant movement and with weaker global guidance. In long-horizon navigation, a strategy that depends primarily on distance is prone to local minima and inefficient detours, so the frontier score remains important for stable and semantically informed exploration.

The **w/o Memory** condition shows why experience reuse matters. Without the memory module, the model’s CSR decreases and its NE increases, which indicates that it becomes less capable of preserving useful historical cues across sub-tasks. This is consistent with the long-horizon nature of the benchmark: later subtasks often benefit from earlier observations, and without a mechanism for storing and reusing those observations the agent is more likely to repeat earlier errors or rediscover already-known regions. The memory module therefore contributes primarily to sequential coherence and error avoidance across stages.

The **w/o Safety Filter** condition demonstrates that planning quality alone is not enough; the resulting motion must still be physically executable.

When the kinematic safety layer is removed, SR and SPL collapse, because the agent can generate non-navigable or physically invalid coordinates that prevent successful long-horizon completion. This confirms that a navigation system for LHPR-VLN must jointly reason about semantics, topology, and motion feasibility rather than treating them as separate downstream concerns.

Overall, the ablation results show that each module addresses a different bottleneck: the scene graph improves grounding, memory improves sequential reuse, frontier scoring improves exploration quality, and the safety filter guarantees executable motion. The full framework is therefore preferable not because it maximizes any single metric in isolation, but because it produces the most stable trade-off across task completion, path efficiency, and sequential consistency.

7.3 Task Horizon, Spatial Reuse, and the Value of Memory

The dataset statistics in Table 3 help explain why a memory-enabled architecture is particularly suitable for this benchmark. As the number of subtasks increases from 2 to 3 and 4, average task steps decrease from 61.81 to 52.88 and then to 46.39, while reentry rate and area association increase sharply. This suggests that longer tasks are not simply linear extensions of shorter ones. Instead, they often contain stronger spatial reuse, repeated room contexts, and more overlap between sub-task regions.

This is important because it implies that long-horizon navigation is not only a matter of searching more, but also a matter of remembering better. When later subtasks are likely to appear in already visited or semantically related regions, experiential memory becomes a direct performance asset rather than a mere storage mechanism. The agent can reuse earlier exploration outcomes, avoid re-entering dead ends, and exploit repeated spatial structure instead of treating each sub-task as an isolated problem. In this sense, the dataset statistics are not just descriptive; they also validate the architectural choice of combining a hierarchical scene graph with cross-task memory.

The comparison between the 2–3 subtask and 3–4 subtask settings in Table 1 further supports this interpretation. The stronger results in the 3–4 subtask setting on several sub-task metrics are consistent with the idea that information reuse becomes more valuable as task structure becomes richer. Although the full-task SR remains strict, the model benefits from the increased opportunity to exploit contextual overlap across subtasks. This indicates that multi-task navigation is not merely harder because it is longer; it can also become more structured, which allows a memory-based

model to perform better when the instruction sequence contains repeated spatial patterns.

7.4 Semantic Ambiguity and Benchmark Sensitivity

The failure-category distribution in Figure 6 reveals that the hardest targets are concentrated in visually ambiguous, instance-rich, or spatially dependent categories such as cabinet, painting, shelf, plush toy, bag, sink, laptop, coffee mug, vase, table, cup, box, towel, book, and desk. Among these, furniture-like categories such as table and desk are especially challenging because multiple valid instances of the same semantic class may exist in a single scene. A semantically correct navigation to one valid instance can still be marked as failure if the benchmark only matches a single annotated coordinate. This makes the observed failure rate partly a consequence of instance ambiguity rather than purely a model weakness.

The issue is even more pronounced for box and book. These targets are often placed on shelves, in rows, or along other extended support structures, so the valid success region is not naturally well approximated by a single circular point. In such cases, a point-centered success criterion can under-credit semantically correct navigation because the correct target behaves more like a surface or linear region than a single location. A similar limitation can arise for large objects such as bed: if the center of the object or the area within one meter of the center is not practically navigable, then success should arguably be evaluated with respect to the object boundary or a reachable perimeter rather than a fixed central coordinate. This reveals an important limitation of coordinate-based benchmark design for large, repeated, or spatially extended targets.

The chart therefore reflects both genuine model difficulty and benchmark sensitivity. Some failures are due to the challenge of visually grounding small or ambiguous objects under partial observability, while others are caused by the rigidity of the current success criterion. For open-vocabulary indoor navigation, especially with repeated furniture and shelf-mounted objects, future benchmarks may benefit from a more geometry-aware success definition that distinguishes semantic correctness from exact coordinate matching.

8. Conclusions

This research proposes a robust, zero-shot framework for Long-Horizon Vision-Language Navigation (LH-VLN) by integrating a hierarchical 3D scene graph with a vectorized experiential memory engine. By explicitly decoupling spatial mapping from cognitive history and bridging them with a four-dimensional active exploration scoring mechanism, the system effectively mitigates cognitive hallucinations and blind exploration.

9. Limitation and Further Work

Despite the significant improvements in long-horizon sequential reasoning, the proposed framework exhibits several limitations that point toward necessary future research directions.

9.1 Perceptual Constraints and Upstream Dependencies

To align with the constraints of low-cost hardware deployments, the framework operates under a strict single front-facing view. While the system implements a rotational visual sweep upon reaching the target neighborhood to compensate for this restricted field of view, it lacks continuous active camera control. This fixed lateral perspective complicates micro-kinematic alignment, limiting the end-to-end Success Rate (SR). Additionally, the system relies heavily on the accuracy of upstream instance segmentation. Traditional computer vision challenges, such as severe occlusions, dynamic lighting variations, and the presence of transparent surfaces (e.g., glass doors), can degrade metric mapping and propagate hallucinated topological nodes into the planning layer. Future work should introduce Active Perception policies that allow the agent to autonomously adjust camera pitch and yaw for dense local reconstruction upon reaching the target vicinity, while integrating uncertainty-aware perception modules to handle visual noise.

9.2 Cognitive-Kinematic Decoupling Bottlenecks

The current architecture employs a discrete modular pipeline where the high-level Vision-Language Model (VLM) issues symbolic commands (e.g., `move_to`) and the low-level planner executes kinematic routing. When encountering tightly clustered obstacles or narrow interaction zones (e.g., a bag hidden under a cabinet), this complete decoupling prevents the low-level controller from feeding specific geometric failure causes back to the semantic planner. This communication bottleneck occasionally leads to local action oscillations and timeouts. Furthermore, the kinematic safety filters are currently optimized for single-story planar structures and cannot facilitate multi-floor navigation (e.g., traversing stairs). To bridge this gap, future research should explore end-to-end Vision-Language-Action (VLA) models that ingest 3D scene graph topologies directly as continuous token prompts, enabling tighter coupling between high-level reasoning and agile obstacle avoidance.

9.3 The Sim-to-Real Gap and Dynamic Environments

The hierarchical 3D scene graph is incrementally constructed under a static environment assumption, treating large furniture and small manipulable objects as immovable topological entities. However, in unstructured real-world scenarios, environments are highly dynamic;

objects may be relocated, occluded, or removed by human intervention. The current system lacks a temporal consistency mechanism to revise graph beliefs in real time. Deploying this system in physical environments will require Lifelong Dynamic Scene Graph Maintenance, enabling the agent to autonomously prune and refresh altered topological branches via active visual comparison upon re-entering a previously visited room.

9.4 Simulation Constraints and Hardware Deployment

Due to the strict temporal constraints inherent to a graduation project, the empirical evaluation was restricted to the Habitat simulation environment. Consequently, there was insufficient time to complete the full reproduction and comprehensive benchmarking of additional external baseline methods. Furthermore, transitioning this framework from simulation to physical robotic platforms remains a critical next step. Future hardware deployment will be conducted in conjunction with Yiwei Xu's graduation thesis, focusing on deploying and fusing these algorithms on a physical Automated Guided Vehicle (AGV) platform. Architecturally, this joint system will utilize a multimodal interface combining voice inputs for natural language task assignment with fused RGB and LiDAR streams for real-time long-sequence navigation and automatic human following. Most importantly, the volumetric TSDF maps and hierarchical 3D scene graphs generated by our framework will be shared as the foundational spatial substrate to support the AGV's local path planning, effectively bridging the gap between static semantic navigation and dynamic human-robot interaction.

9.5 Evolution of Evaluation Protocols

As noted in the discussion of instance ambiguity and spatially extended targets, rigid coordinate-based success metrics inherently penalize semantically correct navigation. While this is a structural limitation of the external benchmark rather than the proposed algorithm, it remains a critical hurdle for the advancement of the field. Future benchmarking frameworks are strongly recommended to adopt instance-agnostic and boundary-aware evaluation protocols to better align geometric success definitions with human-intended semantic logic.

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Appendices

Appendix A: Project supervision forms and project schedule (Gantt chart)

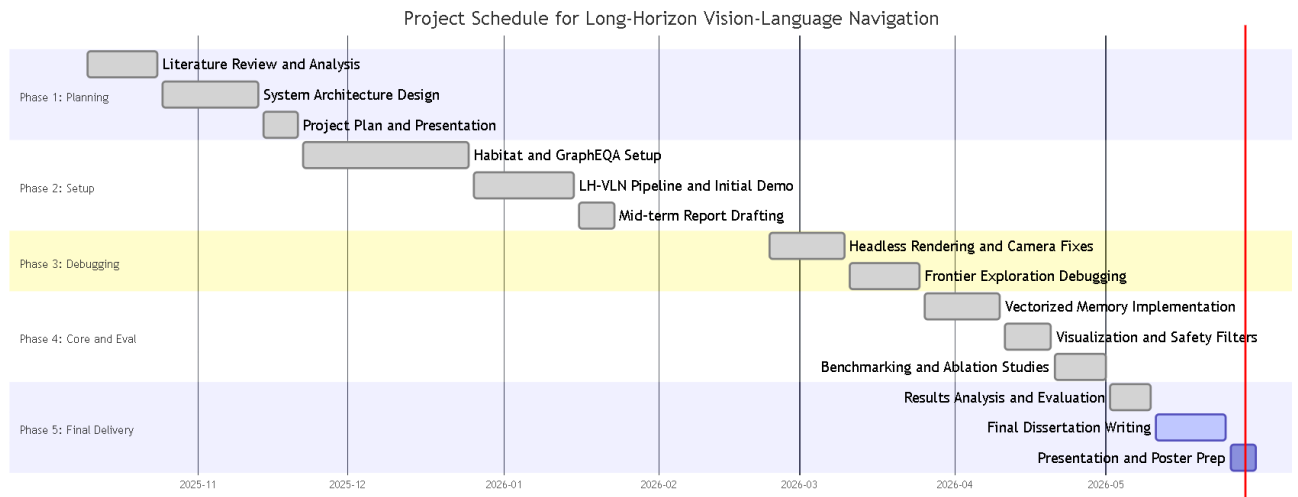


Figure A.1 Gantt chart and detail timelines for this work



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DEPARTMENT OF MECHANICAL, MATERIALS AND MANUFACTURING ENGINEERING

MMME3023 Individual Project Supervision Form

*A minimum of ten formal meetings between students and their supervisors must take place.
For each of these sessions a supervision form must be completed.*

Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 25th Feb

Progress since last meeting (to be completed by the student before the meeting):

I have reviewed the feedback from the mid-term report. Currently, I am fixing the core bugs identified during the initial pipeline test. I encountered significant rendering errors when using Open3D on the headless server, which caused the simulation to crash during visualization.

I believe that my progress has been: Adequate

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

The supervisor suggested abandoning real-time GUI rendering on the headless server. I will modify the script to silently save RGB/Depth frames at each step and compile them into an MP4 video offline for trajectory debugging.

Signed: _____ (Supervisor)

Date of next meeting:

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Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 4th Mar

Progress since last meeting (to be completed by the student before the meeting):

The headless video saving mechanism is now fully functional. I also identified an issue where the agent gets stuck at room entrances. I started tuning the inflation_radius parameter in Habitat (currently testing 0.1) to fix navigable area restrictions.

I believe that my progress has been: Good

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

I need to build a 2D top-down map visualization tool to better track the agent's path and understand why it refuses to pass certain doors. Next week, I will implement the trajectory generation on the 2D map.

Signed: _____ (Supervisor)

Date of next meeting:

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Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 11th Mar

Progress since last meeting (to be completed by the student before the meeting):

I successfully implemented the 2D top-down map with trajectory lines and point cloud saving scripts. I also fixed a critical camera offset bug that was causing the agent's line of sight to clip through objects during close-range interactions.

I believe that my progress has been: Good

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

Begin developing the 4D frontier scoring logic. Since random exploration is inefficient, I need to figure out a way to assign "exploration degree" attributes to the 3D scene graph's region nodes to guide the LLM.

Signed: _____ (Supervisor)

Date of next meeting:

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Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 18th Mar

Progress since last meeting (to be completed by the student before the meeting):

The exploration degree has been successfully attached to the region nodes. I also fixed a logic bug where the agent would repeatedly explore the same frontier because the frontier point wasn't properly cleared from the unvisited list after arrival.

I believe that my progress has been: Good

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

Start the development of the Vectorized Experience Memory module. I will use OpenAI's embedding API to store historical heuristics so the agent can perform cross-task reasoning.

Signed: _____ (Supervisor)

Date of next meeting:

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Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 25th Mar

Progress since last meeting (to be completed by the student before the meeting):

Progress since last meeting: The experience memory module is integrated. However, I found a bug in the task manager: when a single subtask triggers a "give up" command, it incorrectly terminates the entire global episode. I have fixed this so the agent simply skips to the next subtask.

I believe that my progress has been: Good

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

The system needs better traceability. I will enhance the logging system to record the VLM's confidence level and explicitly display the subtask status and the "move_to" target point on the 2D map.

Signed: _____ (Supervisor)

Date of next meeting:

Please keep the completed, signed form safe. At least ten forms must be submitted as an appendix in the final report.



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Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 1st Apr

Progress since last meeting (to be completed by the student before the meeting):

Progress since last meeting: Comprehensive logging is done. To eliminate blind spots, I added a 360-degree rotation maneuver when the agent arrives at a frontier. This significantly improves object detection. I'm now preparing the scripts for the final benchmark evaluation.

I believe that my progress has been: Good

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

Agreed plan: I will write the Evaluate_SPL&OSR.py script to calculate the fine-grained metrics (SR, OSR, ISR, CSR). The supervisor advised running a small batch test to verify the metric accuracy first.

Signed: _____ (Supervisor)

Date of next meeting:

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*A minimum of ten formal meetings between students and their supervisors must take place.
For each of these sessions a supervision form must be completed.*

Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 8th Apr

Progress since last meeting (to be completed by the student before the meeting):

The evaluation script works. During testing, I noticed that the scene_graph_sim.py struggles with room segmentation because Hydra3D's default clustering parameters are not ideal for all HM3D scenes.

I believe that my progress has been: Adequate

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

I will spend the next week tuning the Hydra segmentation parameters. I will also start running the full 63-task benchmarking and the ablation studies overnight on the server.

Signed: _____ (Supervisor)

Date of next meeting:

Please keep the completed, signed form safe. At least ten forms must be submitted as an appendix in the final report.



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MMME3023 Individual Project Supervision Form

*A minimum of ten formal meetings between students and their supervisors must take place.
For each of these sessions a supervision form must be completed.*

Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 15th Apr

Progress since last meeting (to be completed by the student before the meeting):

Progress since last meeting: All benchmarking runs (including the baseline and 4 ablation variants) are complete. I am currently analyzing the failure cases. I noticed a rigid flaw in the benchmark: it severely penalizes valid navigation to large items (like beds or kitchen counters) or scenes with multiple identical tables because the success threshold is strictly tied to a single center coordinate.

Discussed the details of those experiments and the comparative analysis with other methods, as well as the possibility of using different AI models as planners.

I believe that my progress has been: Excellent

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

This finding is a great point for the "Discussion" chapter. I will compile the quantitative tables and draft the failure analysis section highlighting this benchmark limitation.

For smaller models, we can deploy them locally. For larger ones, we will use API keys to execute the planning tasks.

Signed: _____ (Supervisor)

Date of next meeting:

Please keep the completed, signed form safe. At least ten forms must be submitted as an appendix in the final report.



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DEPARTMENT OF MECHANICAL, MATERIALS AND MANUFACTURING ENGINEERING

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Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 22th Apr

Progress since last meeting (to be completed by the student before the meeting):

The quantitative results (Table 1 & Table 2) and failure category bar charts are completed. Due to strict time limits, I have decided to deprioritize the VLM visual verification step and real-world deployment, moving them to the "Future Work" section to focus on thesis writing.

I believe that my progress has been: Good

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

Begin drafting the main chapters of the dissertation, focusing particularly on the Methodology and Results sections. Ensure all Python pseudocode and generated figures are properly formatted.

Signed: _____ (Supervisor)

Date of next meeting:

Please keep the completed, signed form safe. At least ten forms must be submitted as an appendix in the final report.



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MMME3023 Individual Project Supervision Form

*A minimum of ten formal meetings between students and their supervisors must take place.
For each of these sessions a supervision form must be completed.*

Full Name of student: Zibo Zheng

Student ID Number: 20515130

Project title: Dual-Channel Memory Architecture via 3D Semantic Scene Graph for Zero-shot Long-Horizon Vision-Language Navigation

Supervisor: Adam Rushworth

Date of meeting: 29th Apr

Progress since last meeting (to be completed by the student before the meeting):

I have completed the first full draft of the dissertation. All 2D topology graphs and 3D hierarchical scene graph visualizations have been inserted into the report. I also structured the GitHub repository for future open-sourcing.

I believe that my progress has been: Excellent

Signed: _____ (Student) Date: _____

Agreed plan (a draft plan can be completed by the student before the meeting, which may be modified as a result of the discussion):

The supervisor will review the draft. I will refine the academic language, double-check all citations, and prepare the slides/poster for the final presentation.


Signed: _____ (Supervisor)

Date of next meeting:

Please keep the completed, signed form safe. At least ten forms must be submitted as an appendix in the final report.

Appendix B: Prompt Engineering Designs

Our framework employs specific Large Language Model prompts for two distinct functional pipelines: high-level semantic reasoning for navigation planning, and low-level scene graph annotation.

B.1 System Prompts

Semantic Analyzer (used in `get_semantic_similarity`):You are a semantic analyzer. Utilize historical experience if provided to adjust your scoring.

Location Extractor (used in `parse_target_room`):You are a precise extractor of location information from text.

B.2 Semantic Similarity & Relation Extraction

- **Batch Evaluation:**

```
1. {exp_context}
2. Compute the semantic similarity between '{target}' and each of the following
   objects: {names_str}.
3. For each, score on a scale of 0 to 1 (1.0 if identical, high if related), judge
   if there is spatial/containment relation (yes/no), and provide explanation.
4. Respond ONLY with JSON list: [{"object": "name1", "similarity_score": float,
   "has_spatial_relation": bool, "explanation": "string"}, ...]
```

- **Single Evaluation:**

```
1. {exp_context}
2. Compute the semantic similarity between '{target}' and '{object_name}' on a scale
   of 0 to 1.
3. - 1.0 if identical.
4. - High if related (e.g., flower and vase = 0.7 if flower is in vase).
5. - Also judge if there is spatial/containment relation (yes/no).
6. Respond ONLY with JSON: {"similarity_score": float, "has_spatial_relation": bool,
   "explanation": "string"}
```

B.3 Instruction Parsing Prompt

- **Target Room Extractor:**

```
1. {exp_context}
2. Compute the semantic similarity between '{target}' and '{object_name}' on a scale
   of 0 to 1.
3. - 1.0 if identical.
4. - High if related (e.g., flower and vase = 0.7 if flower is in vase).
5. - Also judge if there is spatial/containment relation (yes/no).
6. Respond ONLY with JSON: {"similarity_score": float, "has_spatial_relation": bool,
   "explanation": "string"}
```

B.4 Scene Graph Room Labeling Prompt

- **Room Annotation:**

1. Given the list of objects: {object_names}.
2. Which room are these objects most likely found **in**? Keep explanation very brief.
3. Response format: {"explanation": "Brief reasoning", "room": "bedroom"}

Appendix C: Extended Qualitative Results

Success Cases Analysis

Task description: Take the box from the laundry room to the bookshelf in the office, then retrieve the cup from the desk, and find the computer on the instrument room desk.

The system generates comprehensive visual logs. Below is a frame from the execution video, showing the agent's first-person view, top-down map, and current subtask info:



Graph C.1 GIF for navigation progress in Task 7

For this task, the agent's execution trace demonstrated the following:

- **Subtask 1 (Move box)**
- **Subtask 2 (Retrieve cup)**
- **Subtask 3 (Find computer)**

The agent first explored possible frontier obtaining from tsdf.py. And then it observed a box in living room and updated it in 3D scene graph. The semantic similarity between target object and object candidate is very high, therefore, agent came to the position of candidate and mark the current sub-task as successful as shown in Graph C.2.



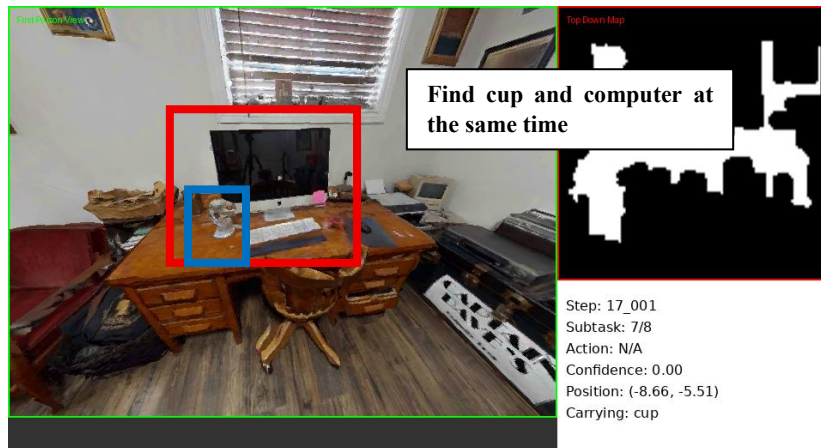
Graph C.2 Subtask 1 Started: Find box

Since no office bookshelf was observed during the previous exploration, the agent continued to explore the area with more abundant information. Finally, at step 9, it saw the bookshelf and reached it at step 10 as shown in Graph 4.



Graph C.3 Subtask 1 Finished: Grabbing box to bookshelf

Since the previous exploration did not find a table with a cup, agent continued the search. Eventually, it found an instrument room (because there's a guitar) and at the same time saw the cup on the table and the computer, completing subtasks 2 and 3.

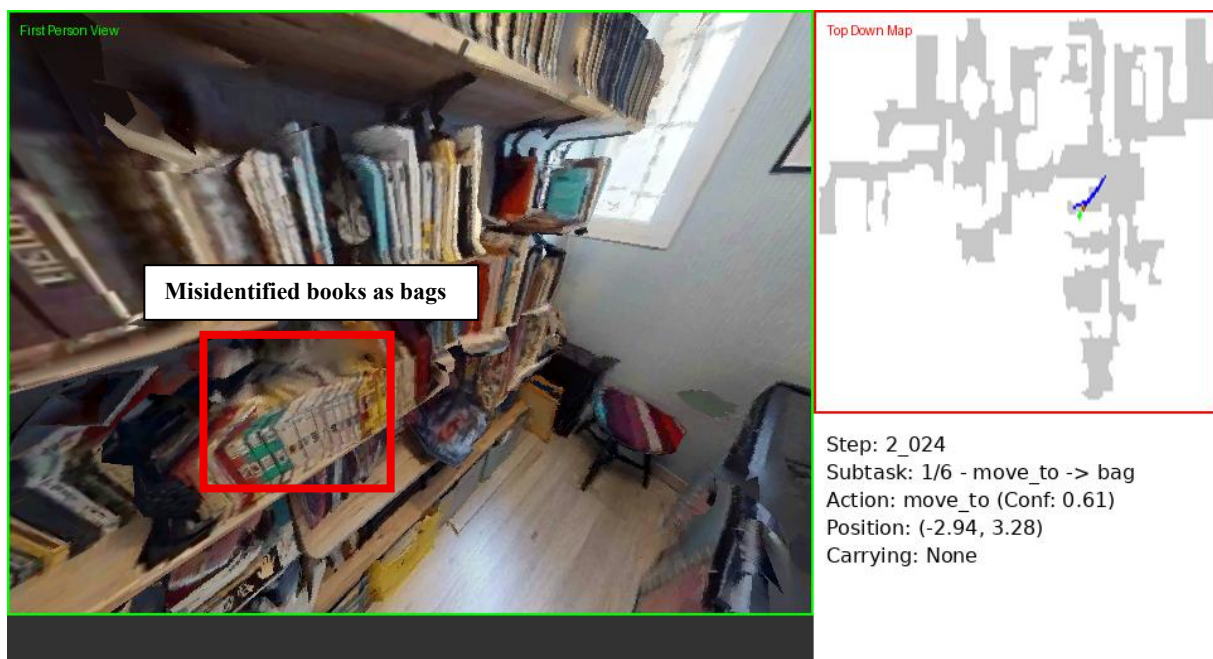


Graph C.4 Subtasks 2 and 3 Finished

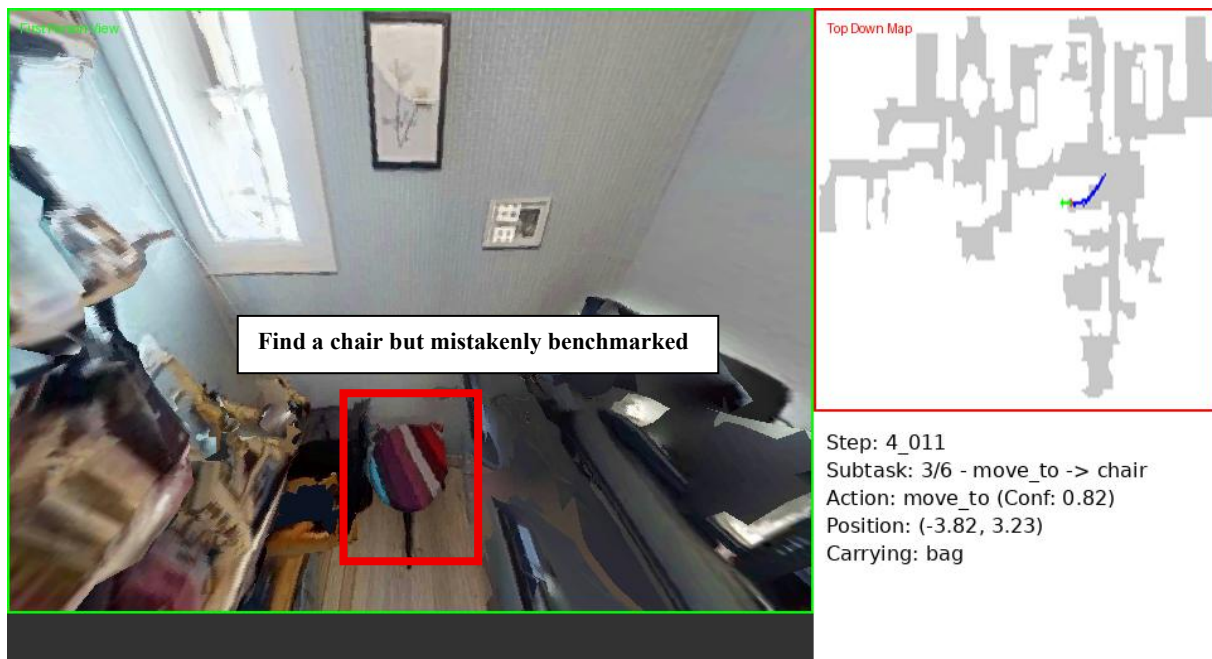
Failure Cases Analysis 1

Task description: Take a bag in office to a chair in office and then retrieve a laptop from the desk.

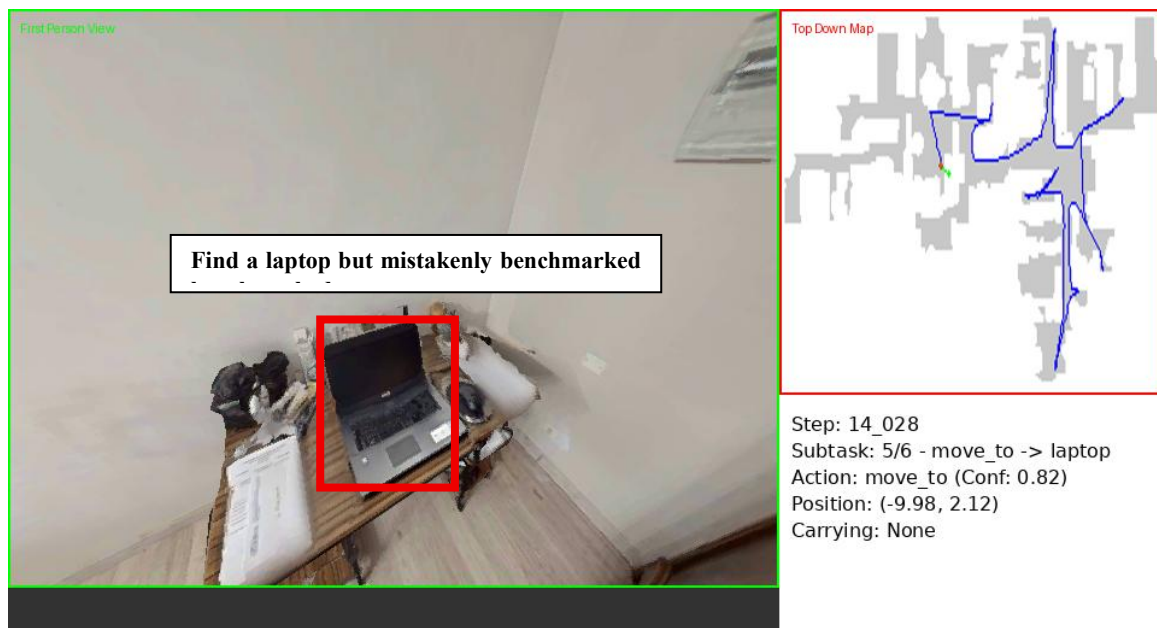
Misidentified the poorly modeled books on the bookshelf as bags:



See the chair beside and move to it. However, it is not the target one and the drawbacks of benchmarking have become apparent.



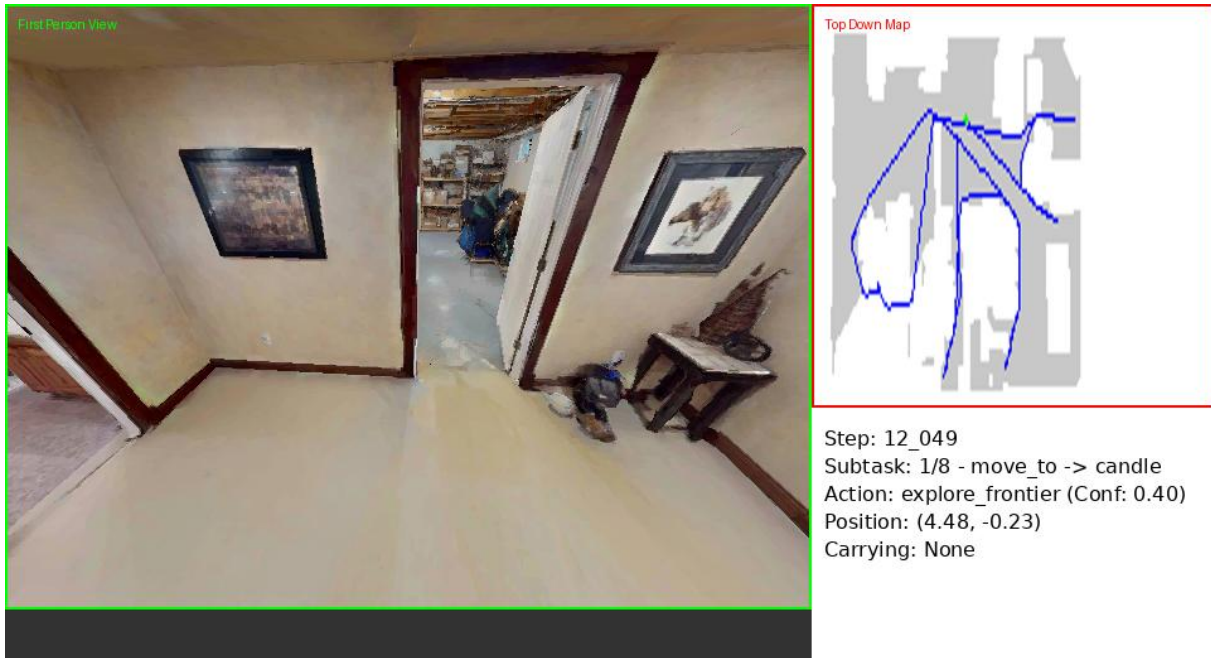
After exploration (trajectory as shown in top-down map), it found a laptop. However, it is also not the target one and this subtask was wrongly marked as unsuccessful.



Failure Cases Analysis 2

Task description: take the candle in bedroom to the wardrobe in office, then take a box in garage to the shelf in living room beside the garage.

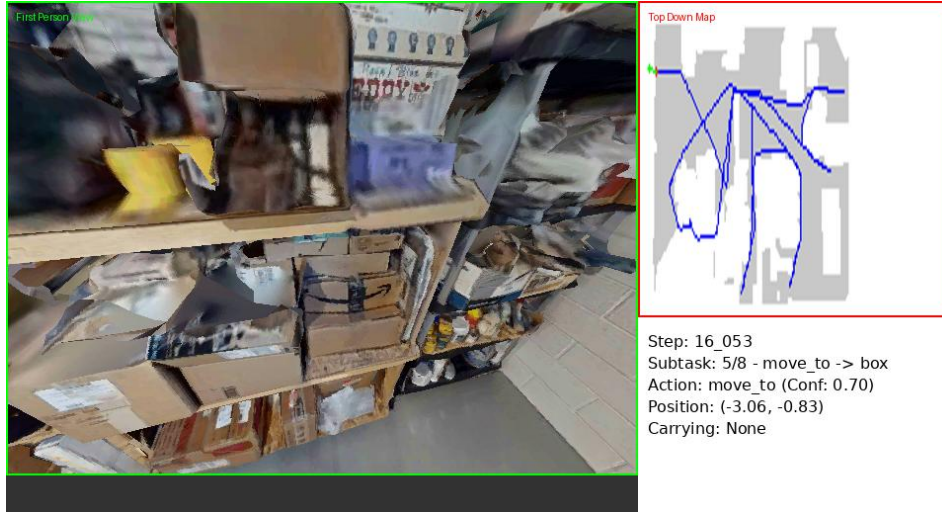
After exploring 12 planning steps, agent found nothing like candle and gave up:



When it came to the next sub-task of "move towardrobe", agent thought he had already seen it during the previous exploration. So agent went straight to what he believed was the wardrobe, but it turned out to be clothes.



Directly move to box built in 3D scene graph memory, however the subtask was marked as fail because there is a whole load of boxes along the wall there and this is not the target one.



The agent did indeed find the shelf in the bathroom, but mistakenly interpreted it as a failure of the sub-task.



Appendix D: Algorithm Pseudocode

This appendix provides the formal pseudocode for the four core operational algorithms of our proposed framework. These algorithms encompass the high-level semantic exploration, low-level kinematic safety, experiential memory management, and topological graph maintenance.

Algorithm 1: Active 4D Frontier Scoring & Gamble Fallback

This algorithm evaluates unvisited frontiers by fusing geometric information gain with semantic potentials. If the scene boundaries are exhausted without a confident target, it triggers a deadlock-prevention gamble fallback.

1. # Algorithm 1: Active 4D Frontier Scoring & Fallback
2. # Input: Target T, Candidate Objects O_all, Unvisited Frontiers F_unvisited, Agent Position P_agent

```

3. # Parameters: Confidence Threshold theta_conf = 0.6, alpha = 3.0, beta = 2.0, gamma_room = 20.0, gamma_obj = 35.0, delta = 1.5
4.
5. Score_best = 0
6. O_best = None
7.
8. for O_i in O_all:
9.     Score_i = SemanticSim(T, O_i)
10.    if Score_i > Score_best:
11.        Score_best = Score_i
12.        O_best = O_i
13.
14. if Score_best >= theta_conf:
15.     return Execute(Move_To(O_best))
16.
17. # Deadlock Prevention & Gamble Fallback Logic
18. if not F_unvisited:
19.     if not FinalCheckDone:
20.         F_unvisited = F_all # Clear blacklist for final check
21.         FinalCheckDone = True
22.     else:
23.         if O_best is not None and Score_best > 0.1:
24.             return Execute(Move_To(O_best)) # Gamble on highest similarity object
25.         else:
26.             return Give_Up
27.
28. # 4D Frontier Evaluation
29. Score_max = -float('inf')
30. F_target = None
31.
32. for F_k in F_unvisited:
33.     Dist = norm(P_agent - P(F_k))
34.     IG = GetInformationGain(F_k) # Calculated via Algorithm 4
35.     UR = GetRoomUnexploredRatio(F_k) # Calculated via Algorithm 4
36.     Sim_room = SemanticSim(T, RoomName(F_k))
37.     Sim_obj_max = max(SemanticSim(T, o) for o in Nearby(F_k))
38.
39.     Score_k = alpha * IG + beta * UR + gamma_room * Sim_room + gamma_obj * Sim_obj_max - delta * Dist
40.
41.     if Score_k > Score_max:
42.         Score_max = Score_k
43.         F_target = F_k
44.
45. return Execute(Explore_Frontier(F_target))

```

Algorithm 2: Kinematic Safety Filter with Linear Backtracking

This algorithm bridges the VLM planner and the physical simulator. It maps symbolic coordinates into strictly navigable mesh paths, avoiding structural collisions.

```

1. # Algorithm 2: Kinematic Safety Filter & Linear Backtracking
2. # Input: VLM Output Target P_target, Agent Position P_agent, Navigational Mesh M_nav
3. # Output: Safe Navigable Target P_safe or Null
4.
5. P_snap = SnapToMesh(P_target, M_nav)
6. if IsNaN(P_snap):
7.     return Null
8.

```

```

9. # Clearance Retraction (30 cm margin)
10. vec = P_snap - P_agent
11. dist = norm(vec)
12.
13. if dist > 0.3:
14.     P_retract = P_snap - (vec / dist) * 0.3
15.     P_safe = SnapToMesh(P_retract, M_nav)
16. else:
17.     P_safe = P_snap
18.
19. # Path Validation
20. if FindPath(P_agent, P_safe, M_nav) is Valid:
21.     return P_safe
22.
23. # Linear Backtracking Fallback
24. for step in range(dist, 0.0, -0.2):
25.     P_test = P_agent + (vec / dist) * step
26.     P_test_snap = SnapToMesh(P_test, M_nav)
27.
28. if FindPath(P_agent, P_test_snap, M_nav) is Valid:
29.     return P_test_snap
30.
31. return Null

```

Algorithm 3: Context-Aware Memory Retrieval and Active Pruning

This algorithm governs the VectorizedExperienceMemory module. It retrieves historical heuristics with a trap-avoidance mechanism and dynamically prunes redundant memories based on an exponential time-decay model.

```

1. # Algorithm 3: Memory Retrieval and Active Pruning
2. # Input: Current Query Q, Current Pos P_curr, Memory Bank M, Max Capacity N_max
3. # Parameters: w_rel = 0.7, w_reach = 0.3, Decay Half-life tau = 600.0
4.
5. # Phase 1: Dual-Dimensional Retrieval & Trap Avoidance
6. V_Q = GetEmbedding(Q)
7. Results = []
8.
9. for E_i in M:
10.     Rel = CosineSim(V_Q, E_i.embedding)
11.     if Rel < 0.3:
12.         continue
13.
14.     Dist = norm(P_curr - E_i.anchor_pos)
15.     Reach = max(0.0, 1.0 - (Dist / 20.0))
16.     Score = w_rel * Rel + w_reach * Reach
17.
18.     # Critical Fallback Trap Avoidance (+0.4 Priority)
19.     if E_i.outcome == "Failure" and 0 <= Dist < 2.5:
20.         Score = Score + 0.4
21.
22.     Results.append((Score, E_i))
23.
24. Sort(Results, by_score="descending")
25. return Results[:top_k]
26.
27. # Phase 2: Active Pruning via Contribution Index
28. if len(M) > N_max:

```

```

29. T_curr = CurrentTime()
30. MinContribution = float('inf')
31. E_victim = None
32.
33. for E_j in M:
34.     TimeElapsed = T_curr - E_j.last_access_time
35.     Recency = exp(-TimeElapsed / tau)
36.     Contribution = (E_j.use_count + 1.0) * Recency
37.
38.     if Contribution < MinContribution:
39.         MinContribution = Contribution
40.         E_victim = E_j
41.
42. Remove(E_victim, from_memory=M)

```

Algorithm 4: Topological Exploration Degree & IG Maintenance

This algorithm runs constantly within the Scene Graph Simulator to dynamically quantify the heuristic Information Gain (IG) of local frontiers and the Unexplored Ratio (UR) of global regions.

```

1. # Algorithm 4: Topological Exploration Maintenance
2. # Input: New Frontiers F_new, Known Regions R_all, Known Objects O_all
3.
4. # Phase 1: Spatial Degree Association (UR updates)
5. for f in F_new:
6.     R_nearest = None
7.     MinDist = float('inf')
8.
9.     for r in R_all:
10.        Dist = norm(P(f) - P(r))
11.        if Dist < MinDist:
12.            MinDist = Dist
13.            R_nearest = r
14.
15.     # If frontier is physically bound to a region (< 5.0m)
16.     if MinDist < 5.0:
17.         R_nearest.exploration_degree += 50.0
18.         Add_Directed_Edge(R_nearest, f, to_scene_graph=True)
19.
20. # Phase 2: Heuristic Information Gain (IG) Calculation
21. def GetInformationGain(Frontier_f):
22.     NearbyCount = 0
23.     for o in O_all:
24.         if norm(P(Frontier_f) - P(o)) < 1.5:
25.             NearbyCount += 1
26.
27.     # Baseline IG is 10.0, penalized by surrounding object density
28.     IG = max(1.0, 10.0 - NearbyCount * 1.5)
29.     return IG

```

Appendix E: Implementation Details & Hyperparameters

To ensure the full reproducibility of our proposed framework, this section details the specific hardware configurations, software environments, and algorithm hyperparameters utilized during the evaluation on the LHPR-VLN benchmark.

E.1 Large Language Model Configurations

The high-level semantic reasoning and planning in our framework are driven by the gpt-4o-mini model provided by OpenAI [38].

- **Model Version:** gpt-4o-mini (API access).
- **Decoding Temperature:** Set to 0.1 across all prompt queries (including semantic similarity scoring, target room extraction, and room labeling) to ensure highly deterministic and reproducible outputs.
- **Output Format:** We enforced strict structured outputs using JSON mode (`response_format=BaseModel`) for all analytical queries, ensuring seamless parsing by the downstream execution pipeline.

E.2 Environment and Simulator Settings

We utilized the Habitat-Sim physics engine to render the HM3D (Habitat-Matterport 3D) dataset. The agent's physical and perceptual constraints emulate a typical low-cost domestic robotic platform.

Table E.1: Habitat Simulator and Agent Configurations

Parameter	Value	Description
scene_type	hm3d	Dataset used for the physical and semantic environments.
camera_type	RGB-D	Single front-facing camera (no panoramic or lateral views).
img_width × img_height	640 × 480	Resolution of the rendered observations.
hfov	120°	Horizontal Field of View for the camera.
camera_height	1.5m	Fixed height of the camera relative to the agent's base.

Parameter	Value	Description
camera_tilt_deg	-30°	Downward pitch angle of the camera.
inflation_radius	0.1m	Navigational mesh inflation for collision avoidance.

E.3 Hierarchical Scene Graph & Perception

The 3D scene graph is incrementally constructed using a TSDF (Truncated Signed Distance Field) planner coupled with Hydra. To select the most representative keyframes for VLM observation, we employ the SigLIP vision-language model.

Table E.2: Scene Graph and Perception Parameters

Parameter	Value	Description
tsdf_grid_size	0.1 m	Voxel resolution for the 3D metric mapping.
frontier_obj_thresh	2.0 m	Maximum radius to associate nearby objects to a frontier.
key_frame_model	SigLIP	Vision-language model used for frame scoring (google/siglip-base-patch16-224).
img_subsample_freq	2	Frequency of subsampling frames to reduce computational load.
topk	2	Number of highest-scoring semantic frames retained per step.

E.4 Navigation and Active Exploration

The active exploration and kinematic execution rely on a set of empirically tuned hyperparameters. These dictate the balance between greedy semantic attraction and global structural exploration.

Table E.3: 4D Scoring and Kinematic Execution Hyperparameters

Parameter	Value	Description
θ_{conf}	0.6	Confidence threshold to trigger a definitive move_to action.
α	3.0	Weight for local Information Gain (IG) in frontier scoring.
β	2.0	Weight for global Room Unexplored Ratio (UR).
γ_{room}	20.0	Weight for macroscopic room-level semantic similarity.
γ_{obj}	35.0	Weight for microscopic object-level semantic similarity.
δ	1.5	Distance penalty coefficient for frontier evaluation.
Safety Retraction	0.3 m	Linear distance the agent retracts from an obstacle mesh.
Backtrack Step	0.2 m	Step size for the linear backtracking fallback algorithm.

E.5 Experience Memory Constraints

The Vectorized Experience Memory engine utilizes OpenAI's text-embedding-3-small for semantic vectorization and maintains hardware viability via an active pruning mechanism.

Table E.4: Episodic Memory Module Hyperparameters

Parameter	Value	Description
similarity_threshold	0.3	Minimum cosine similarity required to retrieve a memory.
max_capacity (N_{max})	50	Maximum number of experience nodes retained in the memory bank.
Decay Half-life (τ)	600.0s	Time constant for the exponential decay of memory contributions.

Parameter	Value	Description
w_{rel} / w_{reach}	0.7 / 0.3	Weight ratio between semantic relevance and physical reachability.
Trap Avoidance Boost	+0.4	Score boost applied to historical failure anchors to prevent deadlocks.
Trap Trigger Distance	< 2.5m	Maximum physical proximity to activate the trap avoidance boost.

E.6 Evaluation Criteria

Following the standard LHPR-VLN benchmark protocol, navigation success is evaluated based on the final physical distance between the agent and the target object's ground-truth coordinate.

- **Strict Success (SR):** The agent must stop within $D_{th} \leq 1.0$ meter of the target.
- **Oracle / Semantic Success (OSR):** The agent successfully routes to the correct spatial vicinity, defined as passing within $D_{th} \leq 3.0$ meters of the target at any point during the trajectory.

Appendix F: Representative Benchmark Task List

To rigorously evaluate the framework's capability in resolving long-horizon dependencies, handling spatial ambiguity, and resisting error accumulation, we curated a subset of 12 highly challenging "hard-case" tasks from the LHPR-VLN benchmark for our ablation studies. These tasks require the agent to navigate across multiple rooms, interact with instance-rich furniture, and memorize previously visited locations. The full natural language instructions for these 12 evaluation tasks are listed below:

1. Move the vase in living room to the cabinet in kitchen & living room, then grab the flower next to it to the sink in bathroom.
2. Take the doll in bedroom and the vase in kitchen to the kitchen countertop item in kitchen in turn.
3. Take the candle in bedroom to the wardrobe in office, then take a box in garage to the shelf in living room beside the garage.
4. Take the book from bedroom to the table in bedroom, then take the blanket in another bedroom to cabinet there.
5. Take the vase in bedroom and the towel in bathroom to the kitchen counter in turn.
6. Move the liquid soap in laundry room to the refrigerator in the kitchen, then retrieve the plate in laundry room to refrigerator.
7. Take the box from the laundry room to the bookshelf in the office, then retrieve the cup from the desk, and find the computer on the instrument room desk.
8. Move the hovering board from bedroom to sink in kitchen, grab the towel in bathroom, and bring the towel to the kitchen sink.
9. Take the towel in bathroom to the kitchen counter in kitchen, then grab the tray on the kitchen island and move it to the table in kitchen.
10. Take the book from the shelf in rec room to the table in office, grab the vase, then move to the coffee table in living room and drop it.
11. Take the dumbbell from living room to the bed in bedroom, then put the telephone on the TV cabinet.
12. Take the vase in the kitchen to the desk in the office, then retrieve the book from the bookshelf in showroom.

Appendix G: Code Repository and Organization

To facilitate future research, reproducibility, and potential hardware deployment integrations, the complete source code, deployment scripts, and configuration files of this framework will be made publicly available.

The repository will be hosted and maintained at our project page:

DCMALH-VLN.github.io.

The core architecture of our modular pipeline is organized as follows:

/workspace/

```
├── cfg/                                # Configuration files for ablation studies
|   ├── lhpr_vln_verify.yaml           # Full framework configuration (Ours)
|   ├── wo_frontier_score.yaml         # Ablation: w/o 4D Frontier Scoring
|   ├── wo_memory.yaml                # Ablation: w/o Episodic Memory
|   ├── wo_safety_filter.yaml         # Ablation: w/o Safety Kinematic
Filter
|   ├── wo_scene_graph.yaml           # Ablation: w/o 3D Scene Graph
|   └── baseline.yaml                 # Baseline configuration
├── graph_eqa/                         # Core algorithmic engine
|   ├── planners/
|   |   ├── vlm_planner_lhpr_vln_gpt.py # Semantic planner for Full
Framework
|   |   └── Ablation_study_planner.py   # Specialized planner for ablation
variants
|   ├── scene_graph/
|   |   └── scene_graph_sim.py         # Topological graph maintenance
|   ├── experience_memory.py          # Vectorized memory engine
|   └── ...                            # Other utility modules
└── scripts/                          # Execution and analysis scripts
```

- | └─ viewer.py # Main runner for the Full Framework
- | └─ Ablation_study_runner.py # Automated runner for ablation studies
- | └─ run_ablation.sh # Shell script for batch experiment management
- | └─ Evaluate_SPL&OSR.py # Quantitative metric calculation script
- | └─ draw_2d_topology.py # 2D hierarchical topology visualization
- | └─ render_hierarchical_dsg.py # 3D Scene Graph + Point cloud visualization
- └─ outputs/ # Evaluation logs and data artifacts
 - └─ lhpr_vln_verification_ours/ # Main benchmark evaluation (Full)
 - └─ lhpr_vln_verification_baseline/ #Ablation study result
 - └─ lhpr_vln_verification_wo_memory/ #Ablation study result
 - └─ lhpr_vln_verification_wo_frontier_score/ #Ablation study result
 - └─ lhpr_vln_verification_wo_Safety_Filter/ #Ablation study result
 - └─ lhpr_vln_verification_wo_scene_graph/ #Ablation study result
 - └─ lhpr_vln_verification_ab_ours/ #Ablation study result
 - └─ lhpr_vln_verification_old/ #Result in mid-term evaluation

Appendix H: Final Metrics Evaluation Results For our work

```
=====
🏆 最终评估成绩单 (Final Metrics) 🏆
=====
SR (Success Rate): 3.17%
ISR (Independent Success Rate): 42.41%
CSR (Conditional Success Rate): 29.48%
CGT (CSR weighted by Ground Truth): 28.03%
OSR (Oracle Success Rate): 71.43%
SPL (Success weighted by Path Length): 1.88%
NE (Navigation Error): 5.09 meters
=====
```